

Multi-objective outpatient scheduling via learning-based adaptive evolutionary search

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ARTICLE INFO

Keywords:

Outpatient Appointment Scheduling
Optimization under uncertainty
Multi-objective optimization
Machine learning for optimization
Genetic algorithm

ABSTRACT

Outpatient Appointment Scheduling (OAS) often involves repeatedly solving multi-objective optimization problems that differ in their waiting lists but share structural similarities across planning periods. In traditional operations research approaches, instances of computationally complex OAS problems are solved from scratch using metaheuristics, overlooking valuable knowledge embedded in past solutions. In this work, we introduce a learning-based adaptive evolutionary approach that leverages historical solutions to improve the effectiveness of future schedules within a fixed computational budget. As a case study, we address a real-world OAS problem for CT-scan scheduling in emergency departments, where three patient categories (outpatients, inpatients, and emergencies) compete for limited resources. We formulate the problem as a multi-objective stochastic program aiming to minimize outpatient waiting times and inpatient and emergency completion times. To solve it, we use the Non-dominated Sorting Genetic Algorithm II (NSGA-II), enhanced with an adaptive warm-start strategy based on a weighted K-Nearest Neighbors method that incorporates a prediction model and optimal assignment to identify the most similar historical instances. Pareto approximations already computed for these instances allow for a similarity-weighted bootstrapping mechanism that generates a tailored initial population. The proposed approach is evaluated in different operational contexts using real-world data from a regional trauma hub in Pavia, Italy.

1. Introduction

Outpatient Appointment Scheduling (OAS) refers to a class of decision-making problems that consist of assigning time slots and resources to patients in ambulatory care settings (Ahmadi-Javid et al., 2017; Cayirli and Veral, 2009). OAS problems are characterized by multiple sources of uncertainty, such as stochastic service times, urgent insertions, and variable demand (Ala and Chen, 2022). In addition, decisions must account for diverse patient characteristics and resource constraints. Given its importance in the management of health services, OAS problems have been widely studied through a variety of modeling approaches, often combined with heuristic and metaheuristic techniques to deal with computational complexity (Castro and Petrovic, 2012; Issabakhsh et al., 2021; Jin et al., 2024).

From an operational perspective, OAS problems are typically recurrent, requiring the periodic construction of appointment plans over successive planning horizons. At each period, the waiting list is updated with new arrivals and resources are allocated according to operational constraints and performance objectives.

Fig. 1 illustrates the decision-making process involved in the general scheduling of appointments for a health service. The OAS generates

an offline schedule that can be adjusted online based on sources of uncertainty. During the execution, new information becomes available as soon as events occur, allowing the schedule to be refined to accommodate uncertain events. Online decisions are made to manage these adjustments, e.g., enabling the prioritization of elective and non-elective patients according to dynamic scheduling rules.

A common OAS goal is to determine the optimal appointment times that minimize interference with emergent and inpatient cases, while also reducing the need for changes to the tentative schedule. This process involves repeatedly solving new instances of the OAS problem, even when they share strong structural similarities with past ones. Traditional approaches, whether exact algorithms or metaheuristics, typically solve each instance from scratch, without leveraging the knowledge embedded in previously obtained solutions. However, the recurring nature of the problem and the structural resemblance across instances open the door to learning from historical instances and their solutions. In particular, information extracted from previously solved instances can be reused to guide and accelerate the resolution of new instances, potentially improving solution quality.

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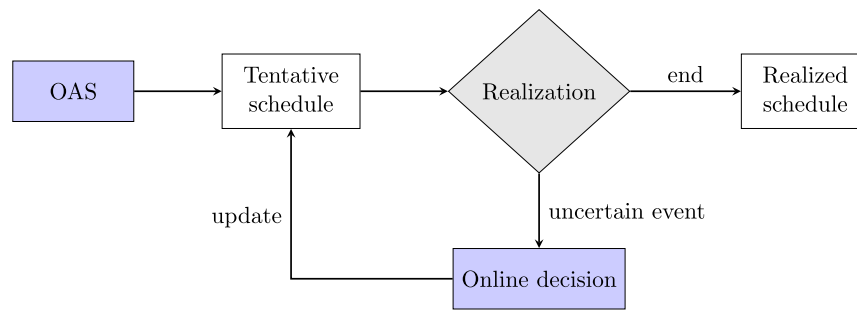


Fig. 1. Flow diagram of the CT-scan management process.

One of the most critical OAS problems concerns Computed Tomography (CT) scan allocation, as delays in access to this key diagnostic resource can lead to longer patient waiting times and increased congestion in the Emergency Department (ED). CT is an advanced diagnostic imaging technique used to visualize a variety of pathological conditions. The increasing demand for diagnostic imaging in Europe is expected to lengthen waiting times, making the management of CT scans and similar services an important issue for healthcare system performance and policy (Siciliani and Verzulli, 2009; Mengou et al., 2021; Biloglav et al., 2020).

In most large European hospitals, emergency radiology has a dedicated section in the ED. Here, the radiology staff interacts with emergency physicians and surgeons in the management of critically ill patients (Scaglione et al., 2021). In the most general operational context, CT resources have to be allocated to perform two activities, namely the execution task and the reporting task, to three different types of patients with different characteristics and needs: outpatients, inpatients, and emergencies (Conlon and Molloy, 2023; Kolisch and Sickinger, 2008).

In this context, outpatients are elective low-urgency patients who require scheduled appointments in advance, typically planned offline weekly. In most emergency radiology organizational settings, these patients are those who have previously been examined during their treatment at the ED and for whom a follow-up exam is scheduled after a few days or weeks. These appointments support the monitoring of patient recovery and the detection of potential complications, contributing to continuity of care (Lee et al., 2019; Ahn et al., 2021). The other two types of patients are non-elective and require dynamic scheduling. Inpatients are patients from hospital departments who need an urgent CT scan during their ward or ED stay, and must undergo the examination within 24 h. Finally, emergency patients are urgent cases from the ED or the intensive care unit that must be examined with high urgency.

This problem presents a managerial challenge due to the need to coordinate the allocation of CT scanners and radiologists while considering multiple criteria. The complexity is further compounded by the wide variety of examinations and sources of uncertainty, including task durations and non-elective requests to be executed in a tight timeframe.

Since OAS involves a wide range of possible objectives depending on the operational context (Golmohammadi, 2021), we focus on the primary patient-centered criteria set by the *Emergency Department of the Radiology Institute of Policlinico San Matteo*, a regional trauma hub located in Pavia, Italy, which inspired our work.

From an outpatient perspective, the primary objective is to minimize waiting time, which is the time elapsed from their scheduled appointment time to the actual execution start. In contrast, report completion time is not considered a critical indicator because outpatients are typically non-urgent cases and do not require immediate clinical action. Logistically, this allows radiologists to complete the reports during less busy periods, and then the reports are delivered after several days.

Contrarily, minimizing the completion time of inpatient and emergency exams is crucial for both clinical and management reasons. A delayed completion for such patients can lead to a late diagnosis of a clinical issue, which may compromise their health status, as well as delay the activities of the department responsible for their care. At the same time, their direct waiting time does not impact their experience since the examination is provided during their stay. In other words, a prompt exam's starting time would merely shift the patient's waiting time to another area of the hospital (e.g., the ED or another ward), while allowing them to proceed with their clinical pathway requires the completion of the report (Duma and Aringhieri, 2023).

In this paper, we formulate the OAS in the context of the emergency CT scan service through a two-stage Stochastic Programming (SP) model that provides an appointment day and time for each patient on the outpatient waiting list. The decision problem considers multiple CT scanners and skilled radiologists as resources, accounting for the main sources of uncertainty: CT-scan execution duration, reporting duration, and emergency and inpatient exam requests. We aim to schedule outpatients in a way that minimizes interference with the execution and reporting of emergency and inpatient exams. Then, dynamic allocation of CT scanners and radiologists is managed by predetermined sequencing rules through the second stage of the SP model.

Since solving the presented SP model with the Sample Average Approximation (SAA) becomes intractable even for small instances and a limited number of scenarios, the adoption of efficient algorithms suitable for stochastic combinatorial optimization becomes necessary (Bianchi et al., 2009). Then, we employ the Non-dominated Sorting Genetic Algorithm II (NSGA-II) (Deb et al., 2002), a widely recognized state-of-the-art approach for solving multi-objective combinatorial optimization problems, due to its robustness and ability to approximate the Pareto front effectively. However, the performance of evolutionary search is highly dependent on the quality of the initial population. To address this, we propose a learning-based adaptive warm-start strategy that takes advantage of structural similarities between historical instances.

The core intuition is that instances with similar structural features (e.g., patient mix, resource load) are likely to share similar Pareto-optimal solutions. Instead of starting the optimization from scratch, we use a K-Nearest Neighbors (K-NN) algorithm to learn Pareto patterns to compute an initial population with diverse high-quality solutions. To this aim, our approach retrieves the most relevant historical instances based on a similarity metric combining assignment-based distance and predicted exam durations. The corresponding Pareto front approximations are then used to generate a tailored initial population through similarity-weighted bootstrapping. This initialization improves both the efficiency and the quality of the search, allowing the algorithm to focus on promising regions of the search space from the beginning, while preserving population diversity compared with random initialization and an alternative rule-based initialization strategy.

In summary, this work provides four main contributions:

- The first stochastic programming formulation of an OAS problem for scheduling follow-up CT exams in emergency radiology;

- The enhancement of multi-objective evolutionary search through a new K-NN-based warm-start algorithm, which involves an optimal assignment in the definition of the similarity among instances;
- The integration of a Machine Learning (ML) regressor for predicting exam durations, enhancing the representation of both patient-specific uncertainty in SP and patient similarity in the warm-start procedure;
- A real-world validation on hospital data, complemented by a generalization analysis demonstrating scalability in settings with a higher outpatient volume.

The paper is organized as follows. Section 2 provides a review of the related literature regarding general OAS problems, CT-scan scheduling, and warm-start strategies for optimization algorithms. Section 3 presents the problem statement and its mathematical formulation through a two-stage SP model. Section 4 describes different strategies to deal with the optimization problem addressed, namely two baseline policies, the SAA, and NSGA-II. Section 5 introduces an initialization approach for the NSGA-II based on current clinical practice and a learning-based warm-start strategy for the NSGA-II. Section 6 presents a computational analysis based on a real case study and an additional scenario to assess the effectiveness of the proposed methodology. Finally, Section 7 discusses the conclusions and suggests directions for future research.

2. Literature review

In this section, we review the application-oriented literature on OAS, with a focus on CT scan scheduling, and the methodological advances in warm-start strategies for optimization.

2.1. Outpatient appointment scheduling

Outpatient appointment decision-making is divided into three categories: strategic, tactical, and operational. Strategic decisions shape the long-term structure of an OAS, tactical decisions focus on medium-term scheduling for patient groups, and operational decisions address short-term patient scheduling (Ahmadi-Javid et al., 2017; Meersman and Maenhout, 2022). According to the classification made in Ala and Chen (2022), decisions at the operational level include the patient selection from a waiting list, their sequence, the determination of an appointment (day and time), patient acceptance or rejection, and resource allocation.

Appointment time decisions can follow either *rule-based* or *optimization-based* approaches. Rule-based approaches rely on predefined scheduling rules and are commonly modeled using queueing systems (Kortbeek et al., 2014; Geng and Xie, 2016) or simulation models (Srinivas and Ravindran, 2018). When system complexity and stochastic interactions limit analytical tractability (e.g., with Markov Decision Processes), Discrete Event Simulation (DES) is typically adopted (Jun et al., 1999). DES also enables the evaluation of online optimization policies that dynamically exploit newly available information (Aringhieri et al., 2020, 2025). Contrarily, mathematical programming is indicated for individual appointment decisions, where optimization is performed under assumptions of deterministic parameters (Agnihotri et al., 2024; Otten et al., 2023) or considering one or more sources of uncertainty, using SP (Bernardelli et al., 2024; Duan et al., 2024; Guo et al., 2024; Wang and Liu, 2022) and Robust Optimization (RO) (Namakshenas et al., 2023; Cheng et al., 2024) frameworks.

Castro and Petrovic (2012) propose a mathematical programming model for radiotherapy pre-treatment scheduling that allocates multiple resources to outpatient activities while minimizing late treatments and overtime, using dispatching rules to generate initial solutions. Begen and Queyranne (2011) develop a multistage SP model for appointment scheduling that accounts for stochastic service times, no-shows,

and emergency arrivals, minimizing idle time and overtime. Chen and Robinson (2014) address an outpatient appointment scheduling problem with shared resources between routine and same-day patients through an SP model solved via Benders decomposition. Issabakhsh et al. (2021) formulate an RO model for chemotherapy appointment scheduling, solved with an adaptive large neighborhood search heuristic. Finally, Mak et al. (2014) propose a distributionally robust optimization model that determines appointment times under partial information on service-time distributions.

An alternative approach to managing uncertainty involves predicting the uncertain parameters of a mathematical program or a simulation model using ML techniques. This framework leverages ML's ability to analyze historical data and identify patterns, enabling more accurate and data-driven predictions of parameter values. In the OAS context, predictive analytics can be used to provide predictions about service times (Golmohammadi et al., 2023; Lai et al., 2024; Srinivas and Ravindran, 2018), to identify patients with high risk of no-show (Srinivas and Ravindran, 2018; Han et al., 2024), to forecast patient demand (Luo et al., 2017; Lin, 2019), and to estimate performance metrics of a given appointment schedule (Wang et al., 2021; Huang, 2015).

Several studies combine prescriptive analytics with ML-based predictions of service times to support appointment scheduling. Chang and Chang (2018) use an Artificial Neural Network (ANN) to predict visit duration in a dental clinic, using patient demographic information and clinical history as features. Then, a DES model is used to evaluate the performance of a scheduling rule informed by the predicted visit duration. Bentayeb et al. (2019) compare linear regression, multivariate adaptive regression splines, decision trees, and ANN for predicting radiotherapy treatment duration. Such information is used to design the appointment slot configuration, which is evaluated considering alternative sequencing rules using simulation. Similarly, Golmohammadi et al. (2023) use simulation to compare different scheduling policies for a university outpatient clinic, which are informed by four different ML methods to define the duration of appointment slots.

While previous works that consider service times as uncertainty focus on decision rules approaches and propose simulation for prescriptive purposes, Han et al. (2024) formalize a multi-appointment scheduling problem for a chemotherapy service through an SP model. They apply ANN to calculate patient no-show probabilities and to predict appointment durations. Both pieces of information are used within the SP model, where the no-show probabilities model the distribution of random variables, while the appointment durations are set as deterministic parameters. A genetic algorithm is then adopted to solve the model.

2.2. CT scan scheduling

Managing a diagnostic facility involves two main decision problems: creating an appointment schedule for outpatients and developing a system of dynamic priority rules to admit patients (e.g., walk-in, emergency, urgent inpatients, etc.) for service in real time (Green et al., 2006).

Previous prescriptive analytics focus mainly on the dynamic scheduling problem. Given a short-term planning horizon, heuristic or optimized sequencing rules to allocate CT scanners to patients are compared through queueing models, where the patient arrival is considered as the only or main source of uncertainty. The OAS problem is addressed by evaluating alternative appointment rules, whose parameters can be optimized through optimization models. Patrick and Puterman (2007) study slot reservation for inpatients, determining the optimal number of reserved slots via direct search to minimize idle time. Kolisch and Sickinger (2008) model CT scanner allocation for inpatients and emergency patients using an MDP with random arrivals and no-shows. Similarly, Gocgun et al. (2011) employ an MDP to derive

optimal policies considering revenue, unexamined patients, and waiting time.

A stochastic optimization problem is defined by Vink et al. (2015) to minimize a convex combination of patient waiting time, idle time, and overtime. Furthermore, an approximation method is used to reduce computational complexity, enabling near-optimal solutions within a reduced running time. In a study by Li et al. (2018), a DES model is proposed to evaluate nine alternative sequencing rules under a fixed outpatient appointment scheme. The exam duration and a possible additional activity for contrast-enhanced scans are considered as sources of uncertainty in addition to the patient arrivals. Different uncertainty aspects are considered in the work of Bhattacharjee and Ray (2018), where walk-in and emergency arrivals, no-shows, stochastic durations, and patient punctuality are modeled in a queueing system. Direct waiting times and machine utilization are the criteria selected to evaluate both appointment and sequencing rules for a single CT scan, using decision trees to estimate different exam durations. Finally, the unique study that considers the allocation of different types of resources is that of Conlon and Molloy (2023), where radiographers and healthcare assistants are included in addition to CT scanners. The evolution of the waiting list length and the utilization of all three types of resources are evaluated for several decision rules by a DES model.

To the best of our knowledge, the OAS for follow-up CT scan exams in an emergency radiology context has never been studied in the literature. In addition, our work is the first approach that provides individualized appointment days and times for CT scan scheduling, taking into account the arrival of inpatients and emergency patients on and throughout a specific weekday. Furthermore, all previous research only takes into account the exam execution task, without considering the report completion time of urgent patients (e.g., cases from the ED), which plays a crucial role in detecting critical issues and reducing the ED length of stay.

2.3. Learning-based warm-start in optimization

In recent years, ML has been increasingly integrated into optimization pipelines to improve solution efficiency. A common approach involves using predictive models to generate warm-starts, enabling solvers to converge faster or explore better regions of the solution space within limited time.

In convex optimization, Sambharya et al. (2023, 2024) propose learning-based warm-start strategies for fixed-point algorithms, including Douglas-Rachford splitting for parametric quadratic programs, with theoretical guarantees. Similar ideas have also been explored in discrete optimization. In this context, Sakaue and Oki (2022, 2023) propose prediction-based frameworks for combinatorial problems such as matching and matroids, improving runtime and addressing cases with multiple optimal solutions.

Another stream of work investigates learning-based initialization in power system optimization. Baker (2019) trains a Random Forest to predict approximate generation and voltage values based only on load data, using historical solutions as training input. These predictions are then used to initialize the solver, reducing computational effort in many cases. Mak et al. (2023) propose a decentralized learning approach in which each agent uses deep neural networks to learn variables that coordinate the system's components. By selecting and training on past optimization runs with good convergence behavior, the method improves efficiency in solving complex energy scheduling problems.

To the best of our knowledge, there is only one previous attempt at warm starting for evolutionary search. In their work, Hassanat et al. (2018) introduce a regression-inspired method to initialize a genetic algorithm for the traveling salesman problem. Their approach recursively partitions the problem into smaller clusters using regression lines and geometric rules, generating structured initial individuals with higher fitness. This technique improves convergence and solution quality compared to traditional initializations.

Unlike prior approaches, our method is the first to leverage historical problem data and solution data to warm-start a multi-objective evolutionary algorithm, specifically the NSGA-II. By learning from previously computed Pareto-approximate solutions, we generate tailored initial populations that enhance search efficiency and solution quality in structurally similar OAS instances. Differently from most learning-based warm-start strategies, which typically predict a single feasible solution or a set of decision variables to initialize deterministic solvers, our approach learns patterns from past Pareto fronts and uses them to construct an initial population that already reflects the structure of high-quality trade-offs. This aspect is particularly relevant in multi-objective optimization, where the quality and diversity of the initial population strongly influence the exploration of the Pareto front.

3. Problem statement and mathematical formulation

From an operational standpoint, a CT scan is divided into two sequential tasks: execution and reporting. During the execution phase, the patient enters the CT scan room, where a radiologist is responsible for reading the medical request and other information related to the patient, while a technician takes care of positioning the patient on the CT scanner and entering data. The image acquisition is then performed solely in the presence of the technician, while reporting is carried out afterward by the radiologist. Typically, there is a one-to-one relationship between the CT scanner and the technician, allowing these two resources to be modeled as a single unit. This unit must be coordinated with the radiologists, who oversee both tasks of the same exam, to ensure continuity of care.

Given a set of outpatients O and a set of days D , we aim to assign each patient an individualized appointment day and time, addressing uncertainty. The main uncertainty factors, represented by the arrival of urgent inpatients and emergencies, and uncertain task durations. Let us indicate with E and I the set of emergency patients and inpatients, respectively. Generic patients are then indicated as $p \in P = O \cup E \cup I$. We do not take into consideration no-shows because in the emergency radiology context, they are largely infrequent.

Another aspect that must be considered is the specialty of the examination. Due to the specific requirements of each exam, not all radiologists are qualified to perform all types of exams. Given a set of radiologists R , we have to assign the patients to radiologists skilled in their specialty. Compatibility between outpatients $o \in O$ and radiologists $r \in R$ is defined by the parameter

$$a_{o,r} = \begin{cases} 1, & \text{if radiologist } r \text{ can be assigned to outpatient } o, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

On the other hand, we assume that all available CT scanners, indicated with the set C , can be used for every exam execution.

In this context, the OAS problem involves scheduling outpatient exams while accounting for the potential arrival of urgent inpatient and emergency requests. The objective is to minimize the direct waiting time of outpatients (i.e., the time between their scheduled slot and the start of the exam) and the completion time of inpatient and emergency cases (i.e., the time between arrival and the radiologist's report completion).

Following the general OAS workflow shown in Fig. 1, the offline outpatient schedule must anticipate that certain online decisions, such as scheduling and sequencing rules, will be applied during execution. This is a critical consideration, as the interplay between incoming urgent requests and these online rules ultimately shapes the realized schedule. Therefore, we assume that outpatient scheduling is planned under the following set of scheduling and sequencing rules, which reflect current clinical practice for managing inpatient and emergency CT scan requests:

- (R1) Emergency CT scans are executed as-soon-as-possible rule when a new patient arrives in the waiting room;

- (R2) Inpatient CT scans are dynamically scheduled with the as-soon-as-possible rule when a new request arises, taking into account the transportation time between their hospital ward and the CT scan room;
- (R3) Independently of the patient types, when allocating the radiologists, CT scan execution has priority over CT-scan reporting, with the second interrupted to perform the first task of a new patient;
- (R4) For a given task (execution or reporting), emergency patients have the maximum priority, followed by that of inpatients, and finally outpatients;
- (R5) Among patients of the same type and requiring the same task (either execution or reporting), a first-come-first-served policy is applied.

We consider discrete time slots $t \in T$ of duration Δ and, according to the case study practice, we assume that outpatients are scheduled in a single work shift (e.g., in the morning). We assume that a radiologist is involved in the execution of the exam for the initial time slots, for a fixed number, while the total execution and reporting durations are expressed as an integer number of time slots.

Outpatient CT scan execution is scheduled within the time slots $t \in T^-$, where $T^- \subseteq T$ is obtained from T by removing a buffer composed of the last time slots of the work shift, in a fixed number, to prevent overtime. However, overtime could be needed for managing urgent inpatient and emergency CT scans. Then, we also introduce the set T^+ , which extends the time slots in $T \subseteq T^+$, for a fixed number, to allow for overtime.

For the sake of simplicity, outpatient CT scan reporting is not explicitly scheduled. The reason is that outpatient reports are not urgent and can be executed in any radiologist's idle time without any impact on the considered performance criteria.

An example of resource allocation for exam execution and reporting, involving two CT scanners (labeled C_1 and C_2) and two radiologists (labeled R_1 and R_2), is shown in Figs. 2–3. The left Gantt charts depict a schedule corresponding to a possible OAS solution: the assigned radiologist is engaged during the initial slot of each exam, while the assigned CT scanner remains occupied for the entire scan duration. The right Gantt charts illustrate the realization of the schedule in a dynamic scenario, following the previously stated scheduling and sequencing rules.

In particular, emergency patient E1 arrives at time 0, when both CT scanners and the compatible radiologist R_1 are available. Thus, C_1 and R_1 are immediately allocated to execute the scan. As soon as the scan is completed, R_1 performs the report writing. This early intervention does not affect the offline schedule, and both outpatients O1 and O2 are examined as planned. At time 20, inpatient I1 arrives; CT scanner C_2 and the compatible radiologist R_2 are available and allocated without waiting time. The report for I1 is initiated by R_2 after the scan. However, a few minutes later, emergency patient E2 arrives. According to rule (R3), the reporting of I1 is interrupted to start the scan of E2, since CT scanner C_1 is available. After the setup phase, R_2 resumes I1's reporting while E2's scan continues on the machine. Once the scan of E2 is completed, R_2 interrupts I1's reporting again to perform the reporting for E2, as established by rule (R4). We remark that all inpatients and emergencies in this scenario are inserted as soon as possible (following rules (R1) and (R2)). Notably, the concurrency situation addressed by rule (R5) does not occur in this example.

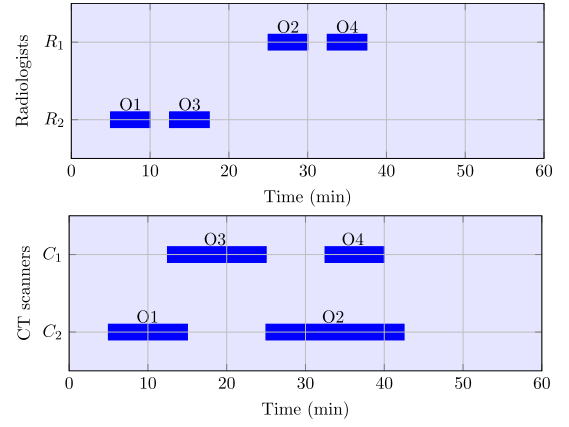


Fig. 2. Example of outpatient exam scheduling with two CT scanners (C_1 and C_2) and two radiologists (R_1 and R_2). Blocks represent the occupation of resources during CT scan executions (blue for outpatients) according to the estimated duration of each exam and are labeled with the patient identifiers O1 – O4. Radiologists and CT scanners are aligned at the beginning of each exam execution.

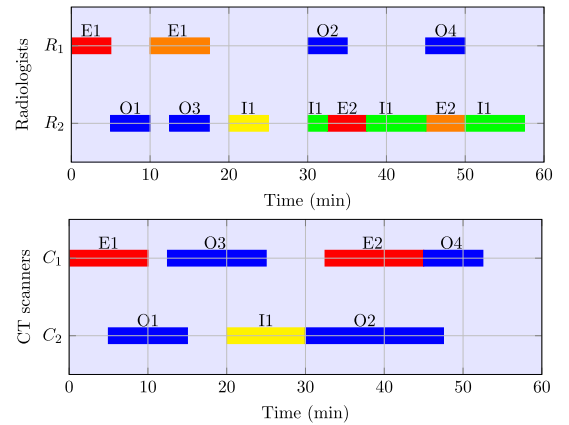


Fig. 3. Example of scenario realization starting from the offline schedule in Fig. 2, where inpatient I1 and emergency patients E1 and E2 arrive at time 20, 0 and 32, respectively. Blocks represent occupation of resources during CT scan executions (blue for outpatients, yellow for inpatients, and red for emergencies) and reporting (green for inpatients and orange for emergencies), performed by radiologists after completion of the scan.

3.1. Sources of uncertainty and random variables

We introduce random variables for the two sources of uncertainty considered in our study, which are modeled through probability distributions that are derived from data. For the sake of simplicity, we introduce the notation by indexing the random variables with a general scenario $\omega \in \Omega$.

The first source of uncertainty is given by the exam requests for inpatients and emergency patients over time, that is, the number of inpatient and emergency arrivals for every time slot $t \in T$ of the planning horizon. Since t is generally a small time interval, we group consecutive time slots $t \in T$ into larger time slots $\tilde{t} \in \tilde{T}$, allowing for a greater statistical robustness and reliability of the distributions. Then, we introduce Poisson random variables modeling the number of non-elective requests in the time slot $\tilde{t}$ of the day

$$e_{\tilde{t},d,\omega} \sim \text{Poisson}(\lambda_{\tilde{t},d}), \quad \tilde{t} \in \tilde{T}, d \in D, \omega \in \Omega, \quad (2)$$

where $\lambda_{\tilde{t},d}$ is the expected number of inpatient and emergency arrivals in the time interval $\tilde{t}$ on day d , representing possible demand fluctuations over both the day and the week, which is typical in EDs (Bell and

Wagner, 2019; Ballarini et al., 2020). We then model the arrival time of each patient p within the aggregated time slot $\tilde{t}$ as being uniformly distributed across the smaller time slots t that compose it. This is equivalent to defining a set of random variables $\epsilon_{\tilde{t},d,\omega}$ representing the arrival time of each non-elective patient p with a discrete uniform distribution over the time slots t that compose $\tilde{t}$, that is,

$$\theta_{p,\omega} \sim \text{Uniform}(\tilde{t}), \quad p \in I \cup E, \omega \in \Omega. \quad (3)$$

Once the arrival of a non-elective patient is determined, the joint assignment of an exam specialty $\psi_{p,\omega}$, patient type $\tilde{\psi}_{p,\omega}$, and urgency code $u_{p,\omega}$ is modeled through the random vector and sampled from a joint categorical distribution

$$\phi_{p,\omega} = (\psi_{p,\omega}, \tilde{\psi}_{p,\omega}, u_{p,\omega}) \sim \text{Categorical}(\rho_{sju}^{(i)}), \quad p \in I \cup E, \omega \in \Omega, \quad (4)$$

where $\rho_{sju}^{(i)} \in [0, 1]$ denotes the joint probability of assigning specialty $s \in S$, patient type $j \in \{\text{inp}, \text{eme}\}$ (i.e., inpatient or emergency), and urgency code $u \in U$ for a patient arriving during the time interval $\tilde{t}$ of a workday. The probabilities satisfy the normalization condition

$$\sum_{s \in S} \sum_{j \in \{\text{inp}, \text{eme}\}} \sum_{u \in U} \rho_{sju}^{(i)} = 1, \quad \forall \tilde{t} \in \tilde{T}. \quad (5)$$

In this formulation, the assignment is carried out in a single step, allowing all possible dependencies between specialty, patient type, and urgency level to be fully captured by the joint probability tensor $(\rho_{sju}^{(i)})$. Such dependencies are important because the flow of inpatient and emergency patients may have different profiles throughout the work shift.

At this point, we are able to split the set of non-elective patients $I \cup E$ into I and E . Given the patient specialty, the following random variable is defined based on the radiologists' skills

$$\alpha_{p,r,\omega} = \begin{cases} 1, & \text{if the radiologist } r \text{ is skilled for the specialty of } p \text{ in scenario } \omega, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

The second source of uncertainty pertains to the duration of the examination and reporting tasks. This uncertainty is addressed using two different approaches, depending on whether the patient is elective or non-elective.

For emergency patients and inpatients, who are not known in advance, we model task durations through probability distributions. The distribution of the exam duration is denoted by $\mathcal{E}_{\phi_{p,\omega}}$ and is understood as the time span between the start and the end of the examination for a patient with profile $\phi_{p,\omega}$. The distribution of the reporting duration is denoted by $\mathcal{R}_{\phi_{p,\omega}}$ and refers to the actual time the reporting radiologist is engaged in preparing the report, excluding possible periods between the start and completion that are spent on other activities. To define the examination and reporting durations of non-elective patients, we use the random variables

$$\zeta_{p,\omega} \sim \mathcal{E}_{\phi_{p,\omega}}, \quad v_{p,\omega} \sim \mathcal{R}_{\phi_{p,\omega}}, \quad p \in I \cup E, \omega \in \Omega. \quad (7)$$

On the other hand, for elective patients that are known in advance, we have specific information at our disposal that can be taken into account to predict the duration of their exams (e.g., exam specialty, type of patient, etc.). In particular, for each outpatient we know a feature vector $\mathbf{f}$, and we can estimate their CT scan execution duration based on these features, that is, we can model

$$\zeta_{o,\omega} | \mathbf{f} \sim \mathcal{E}_{\mathbf{f}}, \quad o \in O, \omega \in \Omega. \quad (8)$$

Finally, we summarize all values taken by these random variables in a scenario with the random vector

$$\xi(\omega) = (\theta_{p,\omega}, \phi_{p,\omega}, \alpha_{p,r,\omega}, \zeta_{p,\omega}, v_{p,\omega}, \zeta_{o,\omega})_{p \in I \cup E, o \in O, r \in R}, \quad \omega \in \Omega. \quad (9)$$

3.2. Decision variables

The decisions of the OAS concern the appointment day and time of each outpatient CT scan, in addition to assigning a radiologist. Let us

introduce the following binary decision variables

$$x_{o,t,d,r} = \begin{cases} 1, & \text{if radiologist } r \text{ is assigned to the exam of} \\ & \text{outpatient } o \text{ in the time slot } t \text{ of day } d, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

These variables are set in the first stage of the proposed SP model; then, a recourse function is computed by the second stage using variables that determine the actual schedule on the basis of realizations of the random variables in scenarios $\omega \in \Omega$. They are defined as follows:

$$w_{o,\omega} \geq 0: \text{ direct waiting time of outpatient } o \in O; \quad (11)$$

$$\ell_{p,\omega} \geq 0: \begin{cases} \text{report completion time of non-elective patient} \\ p \in I \cup E. \end{cases} \quad (12)$$

Further auxiliary decision variables are used in the second-stage model, as reported in [Appendix A](#).

3.3. Two-stage stochastic programming model

We present the multi-objective two-stage stochastic mixed-integer programming model used for the OAS of CT scan exams. The first-stage model is formulated as follows

$$\min \quad \mathbb{E}[Q(\mathbf{x}; \xi(\omega))] \quad (13a)$$

$$\text{s.t.} \quad \sum_{t \in T^-, d \in D, r \in R} x_{o,t,d,r} = 1 \quad \forall o \in O, \quad (13b)$$

$$\sum_{t \in T^-, d \in D} x_{o,t,d,r} \leq a_{o,r} \quad \forall o \in O, \forall r \in R. \quad (13c)$$

In the objective function (13a), $\mathbf{x}$ represents the vector of first-stage decision variables $x_{o,t,d,r}$, while the random vector $\xi(\omega)$ is defined as in Eq. (9). The objective function is the expected value of the recourse function $Q(\mathbf{x}; \xi(\omega))$, which is a vector of two objectives corresponding to the objective function vector of the second-stage model. Constraints (13b) impose that exactly one appointment day, time, and radiologist must be assigned to every outpatient. In addition, constraints (13c) ensure the assignment of radiologists who are skilled in the exam specialty.

It is worth noting that our model allows the assignment of the same resource to overlapping time slots for different patients. While this inevitably results in non-zero waiting time for at least one of the patients involved, such a calculated delay may yield overall benefits for other patients examined on the same day. If these assignments ultimately lead to suboptimal solutions, they will be naturally discarded by the optimization model.

In the second-stage model, we aim to minimize two objective functions, corresponding to the two components of the recourse function, in a given scenario $\omega \in \Omega$

$$Q(\mathbf{x}, \xi(\omega)) = \begin{bmatrix} Q_1(\mathbf{x}, \xi(\omega)) \\ Q_2(\mathbf{x}, \xi(\omega)) \end{bmatrix} = \min \begin{bmatrix} \frac{1}{|O|} \sum_{o \in O} w_{o,\omega} \\ \frac{1}{|I \cup E|} \sum_{p \in I \cup E} \kappa_{p,\omega} (\ell_{p,\omega} - \theta_{p,\omega}) \end{bmatrix} \quad (14)$$

$$\text{s.t.} \quad (\text{A.1})-(\text{A.27}).$$

The first objective, $Q_1(\mathbf{x}, \xi(\omega))$, measures the average direct waiting time for outpatients. The second objective, $Q_2(\mathbf{x}, \xi(\omega))$, captures a weighted sum of completion times for inpatients and emergency patients. Since inpatients and emergency patients differ significantly in urgency, weights $\kappa_{p,\omega}$ are assigned based on their urgency codes $u_{p,\omega}$.

The search space is defined by a set of linear constraints that compute the actual schedule after the realization of the random variables in a general scenario $\omega \in \Omega$. Such constraints are reported and described in [Appendix A](#) and ensure proper task executions and the implementation the decision rules (R1)–(R5).

4. Optimization methods

In this section, we define two uncertainty-unaware baseline policies that allow us to evaluate the contribution of the proposed ML and optimization strategy in comparison with standard fixed rules. Then, before presenting the specific optimization techniques adopted for the optimization problem (13a)–(13c), we describe a strategy for handling uncertainty based on point prediction of outpatient exam durations and Monte Carlo sampling, which is shared by the two proposed frameworks. Finally, the SAA (Kleywegt et al., 2002) and the NSGA-II (Deb et al., 2002) are proposed as uncertainty-aware strategies.

4.1. Baseline policies

We introduce two policies that generalize the current practice observed in the case study and are commonly adopted as baseline approaches in the literature (Cayirli and Veral, 2009; Koeleman and Koole, 2012).

Front-Loaded Wave (FLW) policy Given the total number of outpatients, $|O|$, to be scheduled during the week, we distribute them evenly across the five weekdays, assigning $N_o = |O|/5$ patients per day. If the division is not an integer, we adjust N_o accordingly. Each day is divided into N_a appointment times (waves), starting from the beginning of the shift, with fixed time intervals between consecutive waves. We then assign $\lfloor N_o/N_a \rfloor$ patients to each wave, distributing any remaining patients one by one into the earliest waves of the shift.

Evenly Spaced (ES) policy Given the total number $|O|$ of outpatients to be scheduled during the week, we distribute them evenly across the five working days, assigning $N_o = |O|/5$ patients per day. Within each day, patients are scheduled at regular intervals. Specifically, if N_d denotes the total duration of the daily shift, the interval length is computed as $\lfloor N_d/N_o \rfloor$.

4.2. Uncertainty treatment strategy

As outpatients are known in advance, their examination durations can be estimated through a supervised ML model that provides point predictions. For each outpatient $o \in O$, we compute a deterministic estimate $\hat{\xi}_o$, which is then fixed across all scenarios, i.e., $\xi_{o,\omega} = \hat{\xi}_o$ for all $\omega \in \Omega$. This approach corresponds to a data-driven expected value approximation. Notably, our proposed learning-guided optimization framework is agnostic to the specific method by which the deterministic estimate is obtained.

The arrival process of non-elective patients is generated using the empirical distribution defined in Eqs. (2)–(3), which models the probability of patient arrivals within each aggregated time slot $\tilde{t}$. To preserve the complex dependencies between the patient profile variables $\phi_{p,\omega}$ as described in Eq. (4), the associated weights $k_{p,\omega}$ in the recourse function, and the task durations $\xi_{p,\omega}$ and $v_{p,\omega}$, we adopt a data-driven sampling strategy based on historical records. Once a non-elective patient arrival is generated in an aggregated time slot $\tilde{t}$, we consider the subset of historical non-elective patients who were recorded as arriving within the same time slot. From this subset, we randomly sample one patient with replacement following a bootstrap procedure. The synthetic patient inherits all attributes of the selected historical record. This approach allows us to preserve the empirical joint distribution of variables observed in practice and capture inter-variable correlations.

4.3. Sample average approximation

The SAA method (Kleywegt et al., 2002) is a widely used approach for solving stochastic optimization problems. It approximates the expected value in the objective function by an empirical average over

a finite number of sampled scenarios. In our case, we replace the expectation in Eq. (13a) with a sample average over a finite scenario set Ω' , that is, we solve

$$\min_x \frac{1}{|\Omega'|} \sum_{\omega \in \Omega'} Q(x; \xi(\omega)) \quad \text{s.t. constraints (13b)–(13c)}.$$

Each scenario $\omega \in \Omega'$ is generated according to the uncertainty treatment strategy described above. To approximate the Pareto front from the SAA model, we can adopt the ε -constraint method, a widely used scalarization technique for bi-objective mixed-integer problems. However, due to the combinatorial complexity of the problem and high scenario dimensionality, this approach becomes computationally intractable as the problem size grows. This motivates the use of a metaheuristic approach based on evolutionary search, described in the following section.

4.4. NSGA-II

The NSGA-II (Deb et al., 2002) is an elitist multi-objective genetic algorithm that sorts the population into different levels based on the number of solutions that dominate each one in the Pareto sense. The algorithm begins with an initial population of N candidate solutions. In the baseline setting, each gene of every individual is sampled independently from its feasible domain. At each generation, N solutions are generated through crossover and mutation. We propose tailored crossover and mutation operators designed specifically for our problem. Crossover is performed with probability p_c and selects two parents are sampled uniformly at random from the current population and, for each gene, swaps the genes between the parents with probability p_{cg} to create two new individuals. Mutation occurs with probability p_m per individual, and each gene is mutated with probability p_{mg} to a random feasible value. The current population is evaluated and then merged with the offspring from the previous generation, resulting in $2N$ individuals. Non-dominated sorting is applied to rank the population: the first front contains non-dominated solutions; the second front contains solutions dominated by one other solution; the third front contains solutions dominated by two other solutions; and so on. The top N individuals are selected for the next generation using crowding distance to preserve diversity. This process repeats until a predefined number of offspring is generated or a time limit is reached (Deb et al., 2002).

We propose an encoding tailored to our specific scheduling problem, ensuring that each solution remains feasible with respect to the underlying stochastic programming model. Each chromosome is composed of $2|O|$ genes, where $|O|$ denotes the number of outpatients to be scheduled. The first $|O|$ genes encode the starting time slot of the appointment, with integer values ranging from 1 to $|T^-|$. The remaining $|O|$ genes represent the day assignment for each patient, taking integer values from 1 to $|D|$. The duration of the appointment associated with patient o is equal to ξ_o , which determines the number of consecutive time slots the patient will occupy.

The assigned radiologist is not explicitly encoded but assigned by a routine within the fitness function following a scenario-independent round-robin allocation rule. CT scan allocation is not explicitly encoded but is dynamically assigned, since there is no need to preassign a specific scanner and each patient can be directed to the first available scanner. This design choice allows us to reduce the dimensionality of the solution space in the encoding phase, thereby facilitating the convergence of the evolutionary search.

An illustrative example of the proposed encoding and its translation into a feasible schedule is shown in Fig. 4. The top part of the figure represents a chromosome composed of 12 genes, where each pair of genes encodes the day and starting time slot assigned to a specific patient. The bottom part displays the resulting schedule over five planning days, with each row corresponding to a day and each column to one of the 15 available time slots.

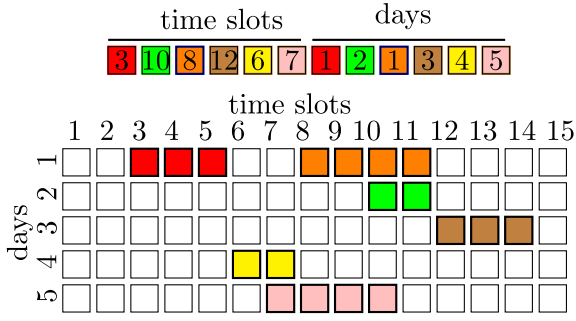


Fig. 4. Toy example of an encoded solution and its corresponding schedule for the OAS problem over a five-day horizon with 15 time slots per day. Six outpatients are considered ($|O| = 6$), each one is represented by a distinct color. For instance, considering the red patient, the chromosome assigns the examination to the third time slot (first gene) on the first day (seventh gene). Given a predicted duration of three time slots, the patient is therefore scheduled from the third through the fifth time slot of the first day.

The fitness function (to be minimized) estimates the objective function of the SP model via Monte Carlo sampling after decoding each chromosome into a complete schedule. A sufficient number of scenarios is produced a priori, then they are used during the entire evolutionary search, allowing for a consistent approximation of the stochastic objective function across different candidate solutions.

Once the outpatient schedule is defined, the algorithms execute Monte Carlo sampling. For each scenario, scheduling and sequencing rules (R1)–(R5) are applied to integrate these additional patients into the existing outpatient schedule, yielding a complete ex-post realization of the planning horizon under uncertainty. Since scenario evaluations are independent, they can be parallelized. Finally, the sample mean of the objective function over all scenarios represents the performance of the chromosome under simulated uncertainty and is used as the fitness score in the evolutionary process. The pseudocode of the proposed algorithm is presented in Appendix C.

5. Knowledge-based initialization approaches

In real-world scenarios, the number of outpatients in emergency radiology may be a small fraction of the patient population. Then, due to the significant uncertainty introduced by the arrival non-elective cases, evaluating the fitness function via Monte Carlo simulation requires a large number of scenarios to ensure stability and robustness. This makes each fitness evaluation computationally expensive. When a reasonable time limit is imposed, this reduces the number of generations that can be executed and may prevent the evolutionary search from reaching near-optimal solutions.

While NSGA-II is effective in exploring complex search spaces, its performance is highly sensitive to the quality of the initial population, especially under limited computational budgets. To address this limitation and enhance the efficiency of the evolutionary search, we propose a rule-based strategy inspired from the clinical practice and a Learning-Based Adaptive (LBA) warm start strategy in which we exploit information from structurally similar historical instances.

5.1. Clinical-practice-based (CPB) initialization

In this section, we describe a rule-based warm-start strategy inspired by scheduling solutions commonly adopted by radiologists. This initialization is introduced as a further benchmark for comparison with the proposed learning-based approach. We assume that patients are uniformly distributed across the week, reflecting current clinical practice. Within each day, examination times are assigned at regular intervals, introducing a predefined slack between consecutive appointments.

For example, assuming O patients per day to be scheduled over T^- time slots, appointments can be spaced with a slack ranging from one slot to $\lceil T^-/O \rceil$. All admissible slack values are considered when generating the appointment schedules; specifically, we construct one schedule for each slack value in this range.

The assignment of patients to specific days is generated randomly, allowing us to produce multiple solutions until the required number for the warm start is obtained. At the same time, the rule-based structure ensures that the resulting solutions remain of good quality and are practically applicable.

5.2. Adaptive learning-based warm start

Let us assume that the scheduling problem has been previously solved over several planning horizons, either in real deployments or through retrospective optimization. For each historical instance h in the set H of the solved instances, we retain an approximation of the Pareto front, represented by the first non-dominated front of the final NSGA-II generation. These solutions form a knowledge base that can be leveraged to warm start the optimization of new instances.

The proposed LBA warm-start framework is composed of four key steps, as illustrated in Fig. 5. Each of these steps is based on well-known and proven algorithms, making the overall approach feasible, reliable, and reproducible for physicians. First, a new instance is processed using a predictive model (e.g., a Multilayer Perceptron regressor) to estimate exam durations for each outpatient. These predictions are then matched against those of historical instances using optimal assignment, enabling a similarity-based ranking. The top K most similar historical instances are selected through a K-NN strategy (Cover and Hart, 1967). From these, we extract a set of solutions that were non-dominated in the corresponding historical instance via similarity-weighted bootstrapping, forming the initial population for the new evolutionary search. This approach ensures that the genetic algorithm begins in a promising region of the solution space to solve the new instance by leveraging structural similarities to past cases.

5.2.1. K-nearest neighbors with distance based on predictions and optimal assignment

The KNN algorithm is a simple yet effective nonparametric classification method that has been widely applied in medical studies, including Rimal et al. (2025), Siva Prasad et al. (2023), Uddin et al. (2022). In our case, to retrieve the most relevant historical cases for a new instance, we compute a similarity metric that combines learning-based duration estimation with optimal assignment, and we use this metric as the distance measure for the KNN algorithm.

We assume that the outpatient service operates under consistent conditions, namely stable resources, policies, and patient flow. Under these assumptions, each instance can be represented by a vector of predicted outpatient durations $\delta = (\delta_1, \dots, \delta_{|O|})$, where each δ_i is estimated using regressor trained on patient-level features (i.e., $\delta_i = \hat{\zeta}_i$). These predictions allow us to identify similar patients from a scheduling perspective. We also assume that the number of outpatients $|O|$ is fixed across instances. If this is not the case, our approach can still be applied by using a padding technique: each duration vector is extended to a fixed length by adding dummy entries (set to zero), which correspond to fictitious patients with a random feasible appointment.

To measure the distance between two instances δ (the new instance) and $\delta^{(h)}$ (a historical instance, $h \in H$), we solve a balanced assignment problem that identifies the optimal one-to-one matching between predicted exam durations. The objective is to minimize the total squared difference across matched pairs. Rather than using a standard Euclidean distance between duration vectors, we adopt this assignment-based approach to account for the inherent symmetry of the scheduling problem: any permutation of the same duration vector yields an equivalent instance. A Euclidean metric would incorrectly

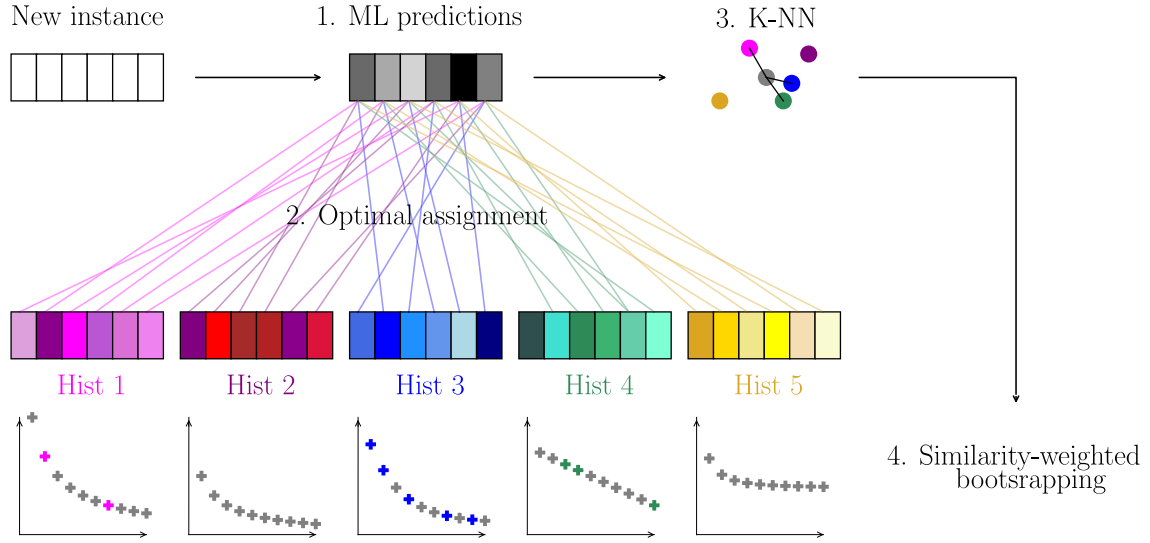


Fig. 5. Framework of the LBA warm start, composed of four main steps. A toy example is illustrated with $|O| = 6$ outpatients, $K = 3$ nearest neighbors, $N = 10$ individuals in the final Pareto front of each instance, and five historical instances (Hist 1–5). Distinct colors identify different instances, while varying shades of each color represent exam durations. In this example, the three closest instances, in increasing order of distance, are the blue, the green, and the pink. We therefore extract five solutions from the Pareto front of the blue instance, three from the green instance, and two from the pink instance, and use them as a warm start for computing the solution of the new instance (in gray).

penalize such permutations, whereas the assignment formulation provides a permutation-invariant measure that reflects the true structural similarity between instances.

Specifically, we define a cost matrix C , whose elements are

$$C_{ij} = (\delta_i - \delta_j^{(h)})^2, \quad \forall i, j = 1, \dots, |O|, \quad (15)$$

and, given the decision variables $X_{ij} \in \{0, 1\}$, $i, j = 1, \dots, |O|$ representing whether the new exam i is matched with the historical exam j , we solve the following assignment problem

$$\text{dist}(\delta, \delta^{(h)}) = \min \sum_{i=1}^{|O|} \sum_{j=1}^{|O|} C_{ij} X_{ij} \quad (16)$$

$$\text{s.t.} \quad \sum_{j=1}^{|O|} X_{ij} = 1, \quad \forall i = 1, \dots, |O|, \quad (17)$$

$$\sum_{i=1}^{|O|} X_{ij} = 1, \quad \forall j = 1, \dots, |O|. \quad (18)$$

This integer linear programming model can be solved in polynomial time using the Hungarian algorithm. The result is an optimal assignment cost that defines the distance between the two instances δ and $\delta^{(h)}$. We compute this cost for each historical instance and apply a K-NN strategy to select the K most similar ones, that is, we choose the K instances $h_1^*, \dots, h_K^*$ with the lowest optimal assignment cost.

It is worth remarking that the ability to identify structurally similar instances hinges on having accurate predictions of outpatient exam durations. In the absence of such estimates, it would be impossible to compare instances meaningfully or transfer knowledge across them. Therefore, the regression model plays a foundational role in the overall approach since its predictive quality directly affects the relevance of retrieved historical cases and, by extension, the effectiveness of the LBA warm-start strategy.

5.2.2. Similarity-weighted bootstrapping

From each of the K selected historical instances, we extract a subset of Pareto-optimal solutions. The number of solutions drawn from each instance is proportional to the inverse of its assignment distance

computed by the optimization model (16)–(18), that is

$$\text{sim}(\delta, \delta^{(h)}) = \frac{1}{\max(\text{dist}(\delta, \delta^{(h)}), \epsilon)}, \quad N_k = \left\lfloor \frac{\text{sim}(\delta, \delta^{(h_k^*)})}{\sum_{k'=1}^K \text{sim}(\delta, \delta^{(h_{k'}^*)})} \cdot N \right\rfloor, \quad (19)$$

where $\epsilon > 0$ is a small number to avoid division by zero (e.g., $\epsilon = 10^{-4}$) and N_k indicates the number of non-dominated solutions drawn from the Pareto front approximation found for the neighbor instances h_k^* , $k = 1, \dots, K$. In this way, the more similar instances contribute more heavily to the warm start. This process is referred to as similarity-weighted bootstrapping.

The selected solutions are then merged to form an initial population of N individuals, which is used to initialize the NSGA-II.

Although the K-NN is a simple and non-parametric method, it serves a crucial purpose in this context by enabling a weighted bootstrapping mechanism. By selecting the top K most similar instances and drawing solutions in proportion to their similarity, we effectively translate instance-level similarity into a structured warm start of the evolutionary search. This hybrid design aims to balance the exploitation of prior knowledge with diversity in the initial population. The pseudocode describing the LBA warm-start initialization is presented in [Appendix C](#).

6. Computational analysis

In this section, we provide a computational analysis by evaluating the impact of the proposed learning-based adaptive evolutionary search under different outpatient volumes. To avoid ambiguity, in what follows, we use the term *scenario* to denote stochastic realizations in Monte Carlo simulations, whereas we refer to the two different patient population cases (real-case and outpatient-mix) as *operational contexts*.

After describing the case study that inspired this work (Section 6.1) and presenting the experimental setup (Section 6.2), we present the analysis of two selected operational contexts:

Real-case context (C_1): we replicate the patient flows and characteristics observed in the dataset from the real-world case study;

Outpatient-mix context (C_2): we increase the proportion of outpatients relative to the total number of patients compared to C_1 , while keeping the overall workload and the ratio between inpatients and emergency cases unchanged.

A scenario analysis is provided to mimic different operational contexts and assess the contribution of the proposed optimization approach under varying conditions (Section 6.3). The *real-case context* provides insights into the potential impact of implementing the approach on the actual case study, while the *outpatient-mix context* reflects a plausible future trend, given the growing emphasis on outpatient services, and allows us to explore contexts where critical resources such as CT scanners are shared not only by outpatient follow-up but they are also used for other elective activities, as seen in different healthcare settings (e.g., operating rooms).

In Section 6.4, we discuss the observed trade-offs and patterns in the extreme solutions, which are those at the boundaries of the Pareto front approximation. In Section 6.5, we compare three different ML methods for predicting exam durations, evaluating them in terms of both predictive accuracy and optimization performance. This comparison serves to motivate the use of the MLP in the first part of the analysis and to assess the robustness of the proposed optimization framework with respect to models exhibiting different levels of predictive performance. Finally, Section 6.6 provides managerial insights.

6.1. Case study

For this computational analysis, we use data of 26,945 CT scans with a total of 49,813 single exams executed between June 2021 and December 2022 by the Radiology Unit of Fondazione IRCCS Policlinico San Matteo, Pavia, Italy. This institution is a hub center for trauma and holds a catchment area of at least 600,000 and up to 1.2 million people, being one of the highest and most complex nodes of the emergency medical system in Italy.

The following features $\mathbf{f}$ are available in the case study's dataset: age, urgency code (*non-urgent*, *intermediate-urgent*, or *highly critical*), sex, mobility status, exams (list of exam procedures performed on the patient), patient type (*outpatient*, *inpatient*, or *emergency*), department of origin, specialty (*Body* or *Neuro*), poly-trauma (binary variable), day type (*weekday* or *holiday*), request date (day and month), request time (hour), number of required examinations, and referring physician. The values of the target variables are obtained from the start and the end timestamps of the exam's execution.

Resources are shared by all three types of patients, i.e., 2.5% outpatients, 23.5% inpatients, and 74.0% emergency patients during the whole day. Two CT scanners are operational during the morning shift (8:00–14:00), which is the one chosen for OAS in the current organizational setting. When restricting the patient population in this shift, we observe 11.7% outpatients, 30.5% inpatients, and 57.8% emergency patients.

The arrival rates of emergency patients and inpatients vary from day to day. When examining individual days, two peaks can generally be observed: one around 9:00 a.m. and another around 11:30 a.m. An exception occurs on Thursday, when only a single peak is observed at 9:00 a.m. The majority of arrivals occur during the first three days of the working week, whereas fewer arrivals are observed on Thursday and Friday. It is worth noting that we account for the transportation time of inpatients; therefore, their arrival at the radiology department is delayed relative to the time their request is made. This delay depends on the distance between the department where the inpatients are located and the radiology department. On the other hand, emergency patients arrive at the radiology department shortly after their requests are made, so we consider their arrival time to be equal to the time of the request.

Two exam specialties are considered: *Body*, which includes examinations of the thorax, abdomen, pelvis, and other soft tissues, and *Neuro*,

which covers scans of the head, spine, and central nervous system. The two specialties share two interchangeable CT scanners, while one skilled radiologist per specialty is available during the morning shift. Furthermore, each patient is assigned an urgency level, with three categories: non-urgent, intermediate-urgent, and highly critical.

Looking at the specialty distribution of emergency patients, 58% are Neuro patients and 42% are Body patients. Regarding urgency categories, 40% of the patients are non-urgent, 53% are intermediate-urgent, and 7% are highly critical. Among inpatients, patients are evenly divided between Body and Neuro specialties, with 54% classified as non-urgent and 46% as intermediate-urgent. Finally, among outpatients, 36% are Neuro patients and 64% are Body patients, with 99% classified as non-urgent and 1% as intermediate-urgent.

The current outpatient organizational practice of our case study consists of scheduling three outpatients every weekday morning shift, resulting in a total of 15 patients per week. In our analysis, we keep this management setting, focusing on the optimization of the OAS in the morning shifts, while the proposed optimization approach revisits both the daily allocation of outpatients and their sequencing within each shift.

6.2. Experimental setup

All experiments were conducted on a Dell OptiPlex 7080 equipped with an Intel(R) Core(TM) i5-10600 CPU @ 3.30 GHz processor and 16.5 GB of RAM. We use Python 3.10 and Gurobi 12.0.0 for the mixed-integer linear reformulation of the SP model (i.e., for the SAA), while for the prediction model of outpatient exam durations we use CatBoostRegressor from the catboost library, and MLPRegressor and DecisionTreeRegressor from the sklearn library. NSGA-II is implemented using the DEAP framework, while its parameter tuning is performed with the Optuna framework.

Data collected between June 2021 and 14 August 2022 has been used to train and validate ML models for predicting outpatient CT-scan execution durations. The same dataset also supports the estimation of empirical distributions to model the arrival patterns and execution durations of the two non-elective patient types within the optimization framework. Subsequently, data from 15 August 2022 to December 2022 (approximately 20 weeks) have been reserved for backtesting the optimization model and evaluating its performance under real-world conditions. We remark that we adopt a chronological train-test split to prevent look-ahead bias, better simulate operational deployment through backtesting, and obtain more reliable performance estimates on unseen future data.

We set 5 min to discretize time into slots and 30 min as the length of aggregated time slots $\tilde{t}$ used to model non-elective patient arrival rates. We assign weights to the second objective of the recourse function (14) according to three different urgency levels identified in the data. Specifically, $\kappa_p = 1/72$ for non-urgent patients, who are not at immediate risk, $\kappa_p = 1/4$ for patients with intermediate urgency who can tolerate brief delays, and $\kappa_p = 1$ for highly critical patients, who require immediate attention. This can be interpreted as 5 min (i.e., one time slot) of prolonged completion time for highly critical patients, corresponding to 20 min for intermediate-urgent patients and 6 h for non-urgent patients, respectively. Such proportions are set according to the nominal maximum waiting thresholds defined for the three types of patients. Furthermore, the parameters $a_{o,r}$ are set to match patients and radiologists according to their specialties. We set the patient transport time parameter $h_{p,\omega}$, defined as the minimum interval that must elapse between the arrival of a patient exam request and its actual start. This value was fixed at 0, 30, or 45 min for inpatients, depending on the physical location of the ward of inpatient i within the city of Pavia, in accordance with the estimated travel time required to reach the CT-scan room. All emergency patients have a null transport time ($h_{p,\omega} = 0$) because the ED is adjacent to the CT scanner room. More details can be found in [Appendix A](#).

Table 1

Comparison of performance indicators for Pareto non-dominated solutions generated by NSGA-II, both with random initialization (baseline), CPB warm start, and the LBA warm start ($K \in \{1, 3, 5, 7, 9\}$), presented for the two operational contexts. The values are the average across the 10 instances. In the last column (W), we indicate with $\uparrow$ if that warm-start configuration is statistically significantly better than the random initialization, with $\downarrow$ if it is worse, and with $\times$ if there is no statistical evidence of a difference. The p -value is reported in parentheses.

	Context C_1 – Real-case							Context C_2 – Outpatient-mix						
	#	C	H	Δ	Sp	MID	W (p -value)	#	C	H	Δ	Sp	MID	W (p -value)
Random	196.1	0.29	13.24	1.22	0.10	2.01	–	45.0	0.05	18.09	1.12	0.34	1.98	–
CPB	191.2	0.21	10.08	1.28	0.02	0.68	$\downarrow$ (0.002)	29.5	0.39	19.09	1.11	0.95	3.51	$\uparrow$ (0.004)
$K = 1$	197.0	0.18	12.52	1.20	0.05	1.82	$\downarrow$ (0.009)	63.6	0.61	19.81	1.10	0.22	1.59	$\uparrow$ (0.027)
$K = 3$	190.2	0.37	13.18	0.19	0.11	1.92	$\times$ (0.556)	82.4	0.64	19.72	1.22	0.49	1.49	$\uparrow$ (0.019)
$K = 5$	197.0	0.49	13.74	1.23	0.12	2.49	$\uparrow$ (0.003)	103.2	0.47	19.46	1.24	0.18	1.84	$\uparrow$ (0.002)
$K = 7$	196.6	0.43	13.71	1.22	0.13	2.14	$\uparrow$ (0.013)	51.6	0.75	19.71	1.13	0.30	1.77	$\uparrow$ (0.002)
$K = 9$	197.0	0.44	13.73	1.21	0.11	2.46	$\uparrow$ (0.013)	106.2	0.63	19.24	1.22	0.33	2.10	$\uparrow$ (0.002)

In all experiments, the fitness function of the NSGA-II approximates the objective function of the SP model through an SAA using $N = 100$ different realizations. Outpatient exam durations are estimated using a ML regression model as discussed in Section 4.2, while emergency and inpatient arrivals and durations are generated through Monte Carlo sampling using empirical distributions. In the real-case context, we set $\lambda_{\bar{s},d} = \lambda_{\bar{s},d}^{\text{real}}$, i.e., the true expected number of inpatient and emergency patient arrivals during time interval $\bar{s}$ on day d . In this setting, we consider three outpatients per day, while the expected number of inpatients and emergency patients is 7.8 and 14.8, respectively. In the outpatient-mix context, $\lambda_{\bar{s},d} = 0.25\lambda_{\bar{s},d}^{\text{real}}$. In this case, we consider 20 outpatients per day, while the expected number of inpatients and emergency patients is 1.5 and 2.7, respectively.

Solving the presented SP model with the SAA method using a general-purpose solver leads to out-of-memory issues. For example, even relatively small instances with only three outpatients, a one-day planning horizon, and 20 scenarios result in memory errors when solved with Gurobi, even for the relaxed problem without the reporting-related constraints (where the objective is to minimize the waiting time for each type of patient). For this reason, we only report results for the baseline configurations and for NSGA-II, where the fitness function is evaluated over 100 possible scenarios of emergency and inpatient arrivals, exam and reporting durations, and real outpatient exam durations.

The parameters of NSGA-II were configured based on the optimal values derived through a parameter optimization process using Optuna: $N = 197$ initial solutions, a crossover probability $p_c = 0.8676$, and a mutation probability $p_m = 0.0324$. The probability of mutation of the single gene and crossover between genes are set equal to 0.5. The number of offspring is set to 92 for the *real-case context*, and to 33 for the *outpatient-mix context*, which results in the same running time of approximately 120 minutes per instance and configuration.

We vary the number of nearest solutions used to generate the initial population of the algorithm, considering values for $K \in \{1, 3, 5, 7, 9\}$. For the LBA warm-start procedure, 100 instances are generated by randomly sampling cases from the training set. Subsequently, NSGA-II is executed with random initialization and the parameters specified above.

For each operational context, we compare performance evaluation metrics on the Pareto front approximations obtained with random initialization, CPB warm start, and LBA warm start across the different values considered. To analyze the quality of the Pareto front approximations, we use several indicators. Given the non-dominated solutions provided by different configurations of the NSGA-II $F_1, \dots, F_m$, we compute their union, $F = \bigcup_i F_i$, and approximate the best possible front F^{ND} by taking the non-dominated solutions in F . The considered indicators are the following

- **Number of Pareto Solutions (#):** the cardinality of F_i ;

- **Coverage (C):** fraction of solutions in $F \setminus F_i$ that are dominated by at least one solution belonging to F_i ;
- **Hypervolume (H):** measures the volume between a reference point r (set equal to the nadir point plus $\epsilon = 0.5$) and the solutions in the front;
- **Diversity (Δ):** measures how evenly and completely the solutions cover the front (Deb et al., 2002);
- **Spacing (Sp):** calculates the deviation of distances between adjacent solutions, evaluating the uniformity of the distribution (Chen et al., 2025);
- **Mean Ideal Distance (MID):** the mean distance from the ideal point (Hosseini et al., 2022);

Indicators #, C, and H have to be maximized, while Δ , Sp, and MID have to be maximized.

Furthermore, we use the Wilcoxon signed-rank test (Wilcoxon, 1945) to compare warm starts and assess the statistical significance. Since the compared warm-start strategies were evaluated on the same instances and no normality assumption could be made due to the limited sample size, the Wilcoxon signed-rank test was adopted as a paired non-parametric statistical test. Where not otherwise specified, comparisons are based on the hypervolume, a widely used metric in the multi-objective optimization literature, as it provides a representative overall assessment. The null hypothesis (H_0) states that the median difference in hypervolume between the compared methods is symmetric about zero, while the alternative hypothesis (H_1) states that the median difference is not symmetric about zero (i.e., paired samples do not come from the same distribution). All tests were conducted with a significance level of $\alpha = 0.05$. The test was conducted in Python using the `wilcoxon` function from the `scipy.stats` module of the SciPy library.

6.3. Analysis of Pareto fronts

In this section, we evaluate the quality of the Pareto fronts using the NSGA-II with different initialization strategies. The historical solutions used to construct the LBA warm start are assumed to be readily available, since the problem is solved repeatedly over time. As a result, the time required to build the warm-start population from this repository is negligible: even in the largest case considered (outpatient-mix context, $K = 9$), it takes less than 5 s.

Table 1 reports the performance indicators described in Section 6.2 for each type of initialization (random, CPB, and LBA). The Wilcoxon signed-rank test is used to compare the CPB warm start and LBA warm start with the random warm start, in order to assess statistical significance with respect to the from-scratch genetic algorithm. Overall, the results confirm the positive impact of LBA warm-start initialization compared to purely random initialization, although the magnitude of the benefit varies depending on the operational context.

In the real-case setting, the LBA warm start with $K \geq 5$ leads to clear improvements in hypervolume (up to 13.74 for $K = 5$), as well as in

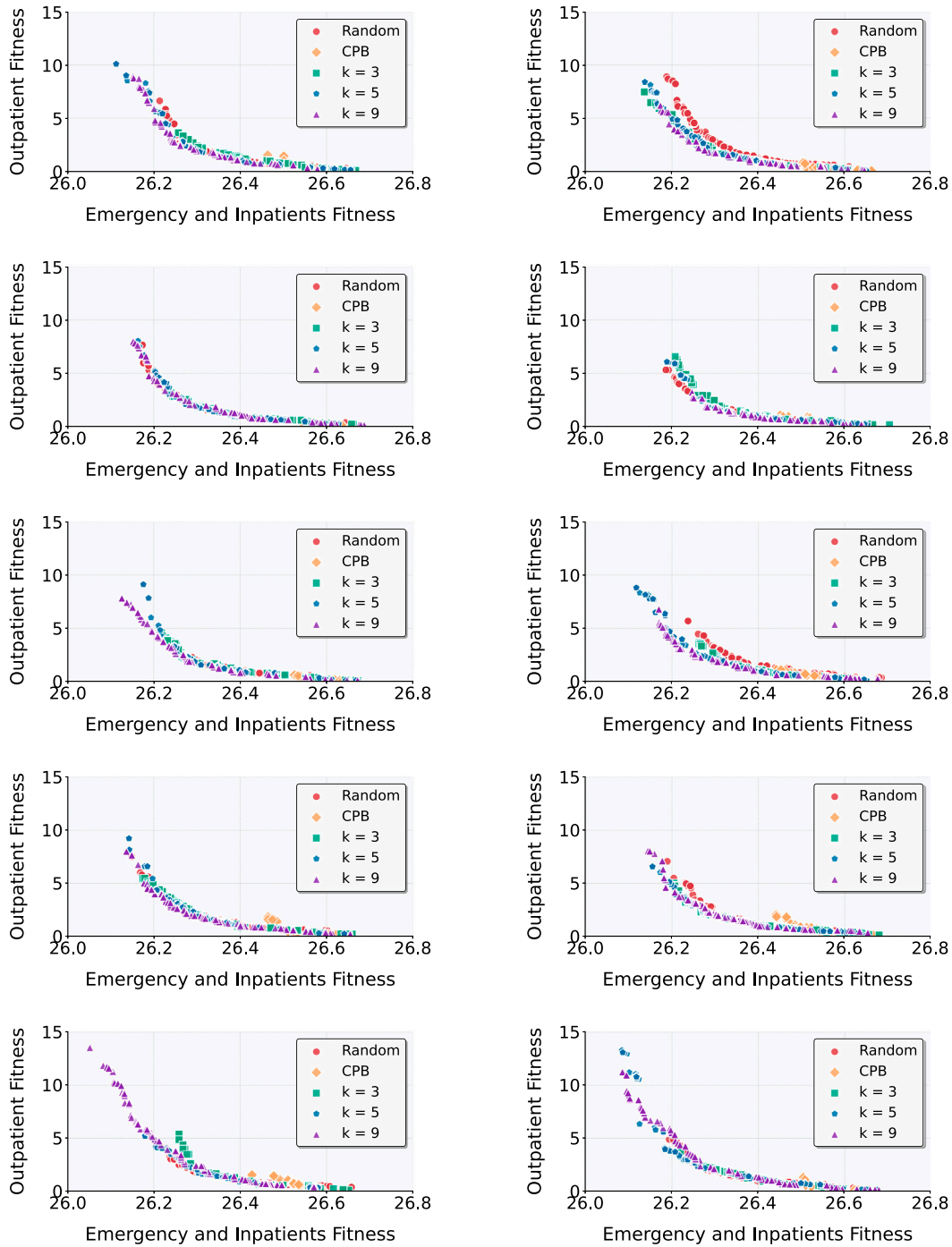


Fig. 6. Context C_1 – Pareto fronts for each validation instance, obtained using random initialization, CPB warm start, and LBA warm start with $K = 3$, $K = 5$, and $K = 9$ in 10 different instances.

most of the other performance indicators: a number of non-dominated solutions equal to the whole population size for $K \in \{1, 5, 7\}$, coverage up to 0.49 for $K = 5$, diversity as low as 1.19 for $K = 3$, and a diversification equal to 8.03 for $K = 9$. The random initialization performs worse across all metrics. The CPB warm start achieves the best results in terms of spacing and MID; however, when considering all metrics jointly, the approach is overall outperformed by the LBA warm start. This is confirmed by the inspection of the Pareto fronts in Fig. 6, where it is shown that the CPB solutions are dominated by the others.

Moreover, the results of the Wilcoxon test show that the CPB warm start and the LBA warm start with $K = 1$ perform significantly

worse than the random initialization. They seem to bias the search too strongly toward the warm-start solutions, ultimately hindering effective exploration of the search space. This suggests overfitting to a specific pattern, leading the evolutionary search to converge prematurely to a suboptimal solution without adequately exploring the search space. For the LBA warm start with $K = 3$, no statistically significant difference is observed, whereas for $K \geq 5$, the results indicate that the LBA warm start performs significantly better than the random initialization.

This behavior reflects the structural nature of the real-case setting: outpatients represent only a small fraction of the overall patient mix, so the degrees of freedom for scheduling are limited, and the large share of inpatients and emergency cases introduces a high level of uncertainty

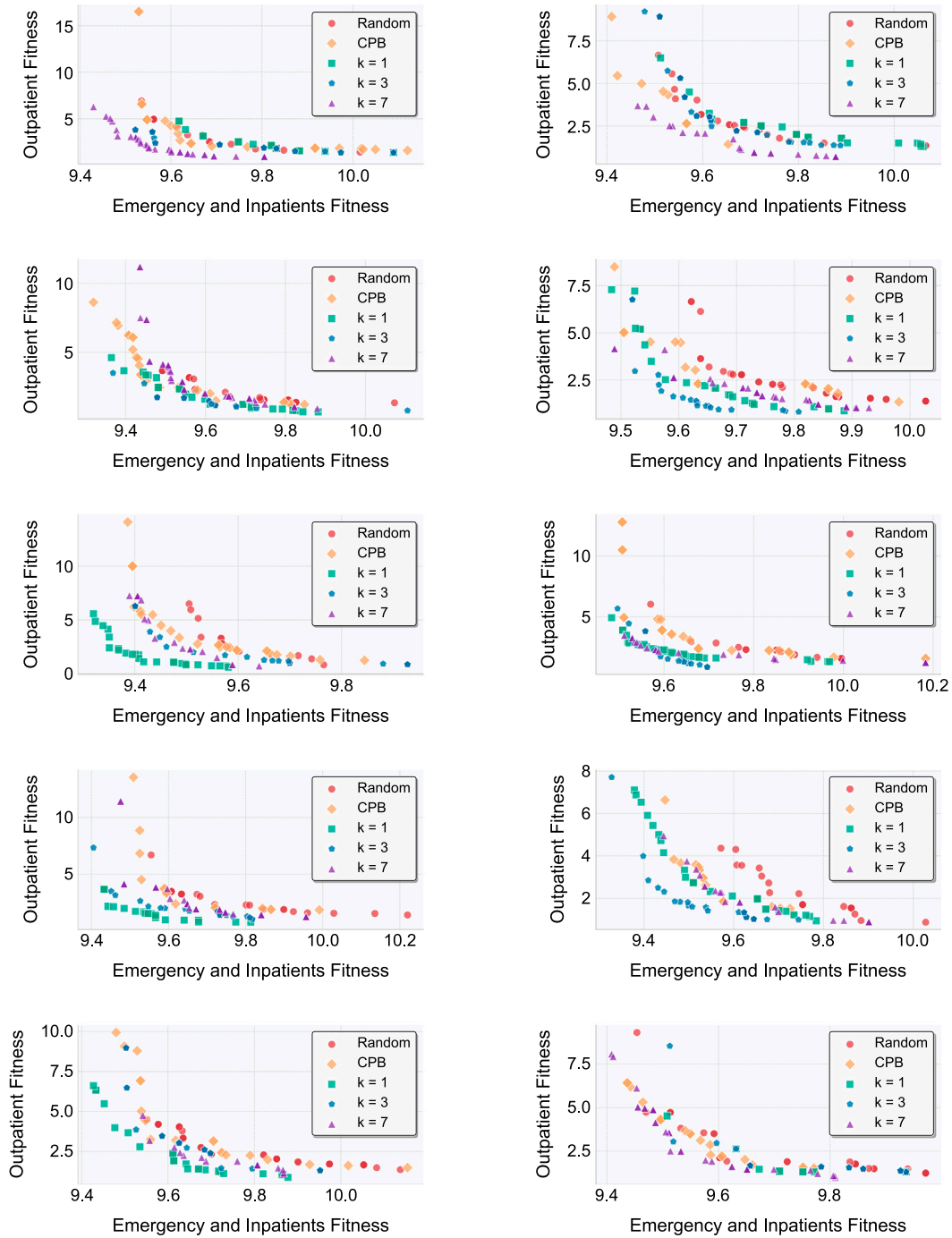


Fig. 7. Context C_2 – Pareto fronts for each validation instance, obtained using random initialization, CPB, and LBA warm starts with $K = 1$, $K = 3$, and $K = 7$ in 10 different instances.

that strongly influences the Pareto front. As a result, any single configuration is able to consistently dominate the others; however, it is necessary to consider more than just the most similar instance for the LBA warm start, as relying on a single instance would be too restrictive. Nevertheless, the LBA warm start approach produces wider and better-distributed fronts, which are more valuable from a decision-making perspective. Even if the CPB warm start achieves better values in terms of spacing and mean ideal distance, it fails to generate a sufficiently wide Pareto front.

This pattern is more evident in Fig. 6, where the Pareto front approximations largely overlap, with a slightly wider separation on the

left-hand side of the front. This suggests that the LBA warm start enables the algorithm to identify better solutions when a higher priority is assigned to non-elective patients, whereas the solutions tend to become increasingly similar as more importance is placed on outpatients. We find statistical evidence that, for sufficiently large values of K (here, $K \geq 5$), the LBA warm start significantly outperforms the random initialization.

In the outpatient-mix context, the number of outpatients increases (i.e., 100 vs. 15) while the overall workload remains constant. Moreover, the size of the solution representation significantly increases: while in the previous context each solution was encoded with 30 genes, in this setting it consists of 200 genes. As a larger proportion of patients

can be scheduled directly and the search space becomes significantly larger and more dispersed, the LBA warm start proves particularly effective. For $K = 3$, the LBA warm start achieves the best mean ideal distance score. For $K = 1$, we obtain the best hypervolume and diversity values. Although the trend observed when varying K shows some variability, the value of the hypervolume for $K = 1$ indicates that this is the best strategy in most cases.

The CPB warm start is outperformed by the LBA approach in every metric. While it generates good extreme solutions, it performs poorly in the central region of the Pareto front. In contrast, random initialization performs poorly overall, highlighting the importance of guiding the search when the decision space expands due to the larger number of elective patients whose scheduling decisions can be made in advance.

Fig. 7 illustrates the Pareto fronts obtained with $K = 1$, $K = 3$, and $K = 7$, together with the front generated by NSGA-II with random initialization and with CPB warm start. When examining the ten instances in this operational context, a clear improvement of the LBA warm start over both random initialization and CPB warm start can be observed, although the best value of K depends on the instance.

From a statistical perspective, we find evidence that all knowledge-based initialization strategies (both the CPB warm start and the LBA warm start for each value of K) perform better than the random initialization. Overall, the proposed approach consistently improves the performance of NSGA-II, although the parameter K needs to be tuned according to the problem characteristics in terms of size and uncertainty. In the operational contexts analyzed, the number of decision variables appears to determine the overall impact of the LBA warm start on solution quality, while the choice of the best value of K seems to be influenced by the level of uncertainty, that is, by the proportion of controllable decision variables to stochastic variables that affect the decisions ex post.

6.4. Analysis of extreme Pareto solutions

In this section, we evaluate the performance of NSGA-II with the LBA warm start by focusing on the two extreme solutions of the Pareto front. These correspond to a clear preference for one of the objectives over the other: minimizing the waiting time of outpatients or minimizing the completion time of inpatients and emergency patients.

All reported results refer to average performance values computed over 10 validation instances. For the backtesting analysis, we reserve the last 10 weeks of collected data, which are not included in the dataset used to train the predictive model, thus allowing us to assess the generalizability of the solutions under realistic, previously unseen conditions.

In the real-case context, the outpatient task durations are replaced with those observed over these 10 weeks. The arrival times of emergency patients and inpatients are sampled from the empirical distributions, while their task durations are generated via bootstrapping, as explained in Section 4.2. This process generates new scenarios compared to those used for evaluating the solutions in the NSGA-II offspring, thereby avoiding information leakage. Conversely, in the outpatient-mix context, we perform bootstrap resampling by randomly selecting (with replacement) a set of 100 patients per week from the pool of outpatients observed in the same period, thus generating 10 realistic instances.

The results are then compared against the baseline policies introduced in Section 4.1, as well as the CPB warm start and the LBA warm start with $K \in \{1, 3, 5, 7, 9\}$, in order to evaluate the potential benefits and limitations of the proposed approach with respect to established scheduling strategies.

Regarding the baselines, in the ES policy, we schedule patients to be evenly spaced during the first 5 h of the shift. For the case study, this corresponds to 3 patients per day: one at the beginning of the shift, the second after 2 h, and the third after 4 h. In the outpatient-mix context,

we schedule 20 patients per day, starting from the beginning of the shift, with 15 min between each pair of patients.

In the FLW policy, for the case study, we adopt the actual policy of the hospital by scheduling 3 patients per day: the first at the beginning of the shift, the second after 30 min, and the third after 1 h. In the outpatient-mix context, the best parameter configuration for this policy results in 20 patients per day, scheduled in 3 waves (7 patients in the first wave, 7 in the second, and 6 in the last), with 30 min between waves.

6.4.1. Real-case context

Fig. 8 shows solutions obtained with the different initialization techniques (Random, CPB, and learning-based adaptive with $K \in \{3, 5, 9\}$), evaluated on the validation instances using the actual exam durations and compared with the baseline policies. Figs. 8(a) and 8(b) report, respectively, the fitness values for non-elective patients and outpatients when considering the extreme solution that minimizes the fitness of non-elective patients.

In Fig. 8(a), the ES policy is the worst-performing method, followed by random initialization, whose performance is affected by prediction errors. The FLW policy achieves reasonable results, suggesting that the current practice at Policlinico San Matteo is effective and comparable to the performance obtained with the extreme solution obtained using the NSGA-II with the CPB warm start. The LBA warm start with $K \in \{3, 5\}$ performs slightly worse than the FLW baseline, whereas with $K = 9$ it outperforms all other methods. These findings highlight the substantial contribution of the proposed LBA warm-start approach, which enables NSGA-II to move from underperforming relative to the baseline policies (i.e., random initialization) to consistently achieving superior results (i.e., LBA warm start with $K = 9$).

Fig. 8(b) shows that when using LBA warm start, prioritizing non-elective patients leads to increased waiting times for outpatients. Except for the FLW policy, which schedules outpatients at the beginning of the shift, and the CPB warm start, all other methods negatively affect outpatient performance. The extreme solutions produced by the LBA warm start, therefore, appear more polarized than those of the FLW policy and CPB warm start. The latter represents a more intermediate trade-off, yielding slightly worse performance for non-elective patients but substantially better outcomes for outpatients.

On the other hand, Figs. 8(c) and 8(d) report, respectively, the fitness values for outpatients and non-elective patients when considering the extreme solution that minimizes the fitness of outpatients. In Fig. 8(c), as in the previous case, the ES policy is the worst-performing method, followed by the random initialization, the LBA warm start with $K \in \{5, 9\}$, and the FLW policy. The LBA warm start with $K = 3$ and the CPB warm start perform best, prioritizing outpatients more effectively than the other methods. Fig. 8(d) shows that prioritizing outpatient waiting times, in most cases, also leads to reduced completion times for emergency and inpatient examinations. This effect may be explained by the priority structure of non-elective exams: outpatients experience shorter waiting times when they arrive during periods with no emergency patients or inpatients requiring service. This scheduling configuration, in turn, preserves additional capacity for non-elective patients.

We perform the Wilcoxon signed-rank test to assess the statistical differences between the LBA warm start with $K = 5$ and the two baseline policies. Since the policies do not provide a hypervolume value (they generate a single solution rather than a set of non-dominated solutions), we instead use the fitness values obtained for non-elective patients (emergency patients and inpatients) and outpatients to perform the test. When compared with the FLW policy, the LBA warm start is not statistically significantly better (p -values of 0.16 for the solution prioritizing outpatients and 0.43 for the solution prioritizing non-elective patients). In contrast, it is statistically significantly better than the ES policy (p -value of 0.002 for both patient categories).

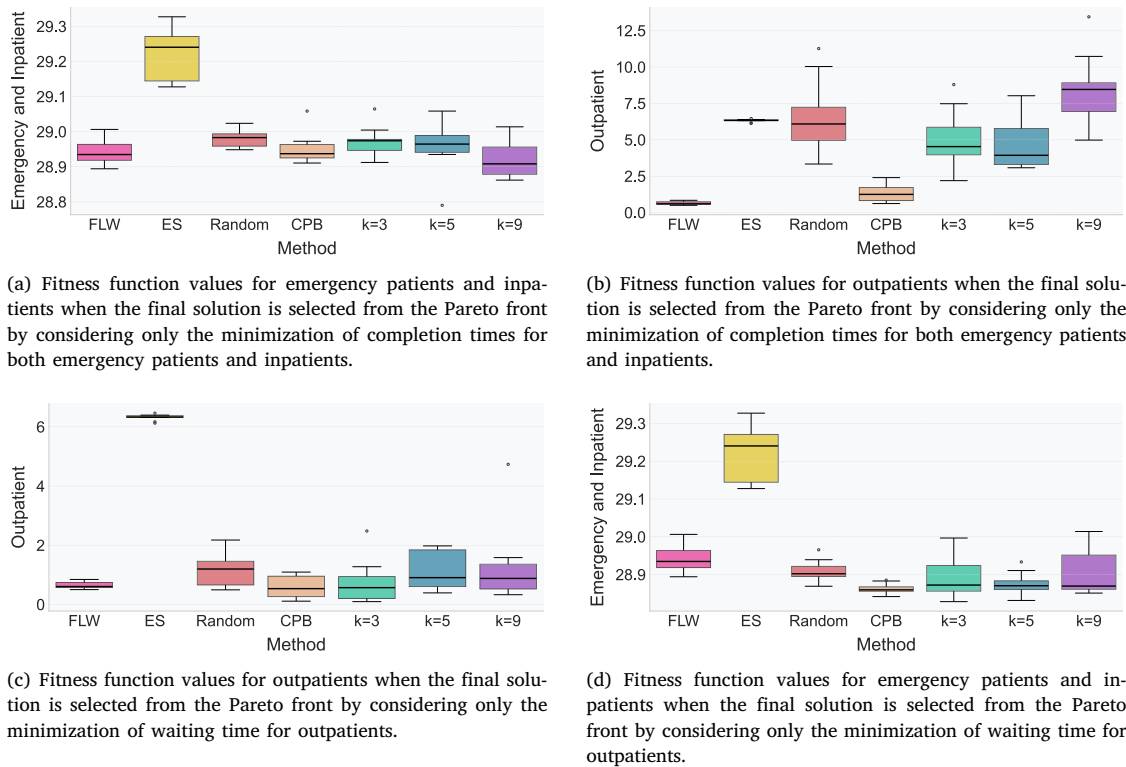


Fig. 8. Comparison of fitness function values for non-elective patients and outpatients under various strategies for final solution selection from the Pareto front.

We can conclude that, although our approach is not explicitly designed to target extreme solutions, but rather to discover well-balanced solutions across the Pareto front, it still achieves performance that is statistically comparable to (and in some cases, better than) the two baselines, even when evaluated on extreme solutions alone. This result highlights the robustness of our method: not only does it provides a diverse set of balanced trade-off solutions that jointly optimize both objectives, but it also find competitive extreme solutions.

To better understand the dynamics behind the extreme solutions provided by the LBA warm start, we present two examples of weekly schedules obtained in one of the instances, with $K = 5$. Note that outpatients are assigned only a day and time for their exam, without specifying a particular CT scanner. Each patient is then scheduled on the first available CT scanner at that time.

In Fig. 9, we report the Gantt chart of a solution that prioritizes the reduction of non-elective patients' waiting times. This solution reveals a pattern in which a large number of outpatients are scheduled in close succession on Thursday at the end of the morning shift. Two factors explain this concentration. Thursday is the day of the week with historically fewer emergency and inpatient arrivals in the final hour, which reduces the likelihood of interference in outpatient appointments. In addition, the stacking and the overlap of outpatient exams reduce the probability that a non-elective patient arrives while CT scanners are occupied with outpatient activity, which is the only circumstance in which a non-elective patient must wait.

This schedule significantly lowers the completion times of non-elective patients, which is a primary objective from a clinical urgency perspective. However, it raises concerns regarding fairness and the quality of service offered to outpatients. For this reason, while valuable as a boundary-case analysis, an implementable solution should be identified at a different point on the Pareto front, which justifies the use of a multi-objective optimization framework. In Fig. 10, we select a solution that prioritizes the reduction of outpatients' waiting times. In the Gantt chart, we observe that the majority of outpatients are

scheduled at the beginning of the shift, before the peak of non-elective patient arrivals. In both schedules (i.e., Figs. 9 and 10), few or no outpatients are assigned on Monday and Wednesday, while the majority are concentrated on Thursday and Friday.

6.4.2. Outpatient-mix context

Fig. 11 shows extreme Pareto-optimal solutions obtained with the different initialization techniques (random, CPB, and LBA with $K \in \{1, 3, 7\}$), evaluated on the validation instances using the actual exam durations and compared with the baseline policies

In Fig. 11(a), when considering the extreme Pareto-front solution that minimizes the weighted completion time of non-elective patients, the LBA warm start enables NSGA-II to further improve its performance compared with random initialization. However, the results remain comparable to those achieved by the CPB warm start and are slightly worse than those obtained with the FLW policy. The effect of these extreme solutions on the average waiting time of outpatients is illustrated in Fig. 11(b). The solutions obtained with the LBA warm start outperform those of the FLW policy and the CPB warm start, indicating that the former dominates the latter. In the same figure, the ES policy achieves significantly lower outpatient waiting times, but at the expense of worse completion times for non-elective patients. Consequently, the overall performance of the ES policy lies in a more interior region of the Pareto front compared with the extreme NSGA-II solutions considered in this analysis.

The other extreme solutions, which assign the greatest importance to the outpatient fitness function, exhibit a different performance hierarchy. As shown in Fig. 11(c), the ES policy achieves lower outpatient waiting times (close to zero) compared with the FLW policy and all initialization configurations considered. However, Fig. 11(d) indicates that the solutions obtained using the CPB warm start, and to an even greater extent those obtained with the LBA warm start for all explored values of K , achieve superior performance for emergency patients and inpatients. In this case, the solutions generated by the ES policy

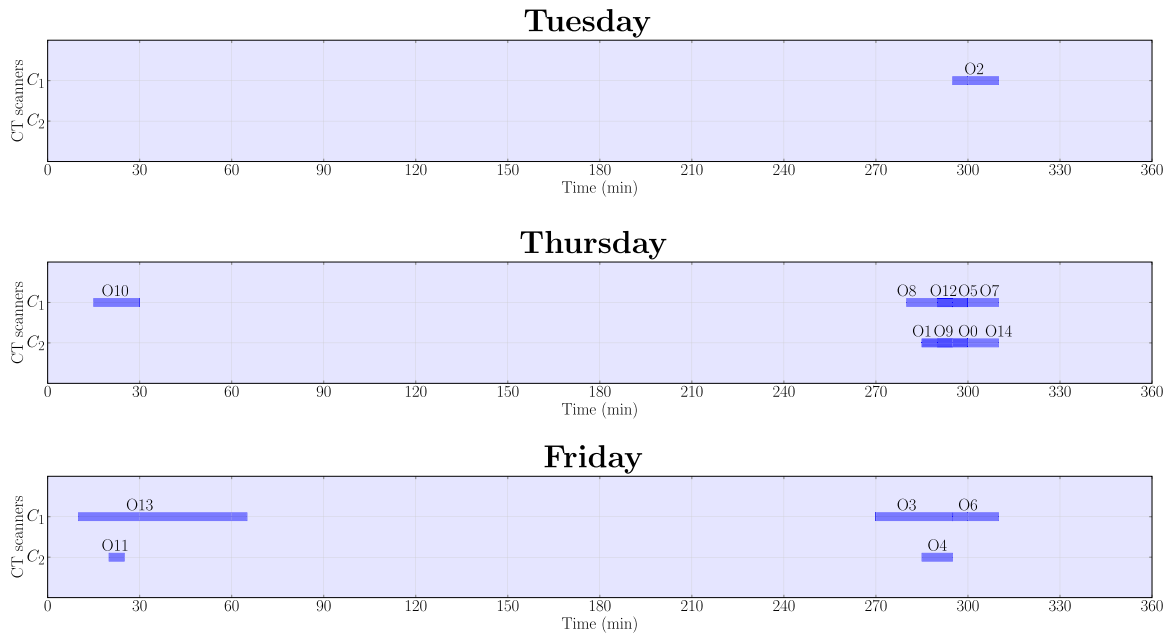


Fig. 9. Context C_1 – Example of schedule generated by NSGA-II with LBA warm start $K = 5$, when the final solution is selected from the Pareto front by considering only the minimization of completion times for both emergency patients and inpatients. We present only the days with scheduled outpatients.

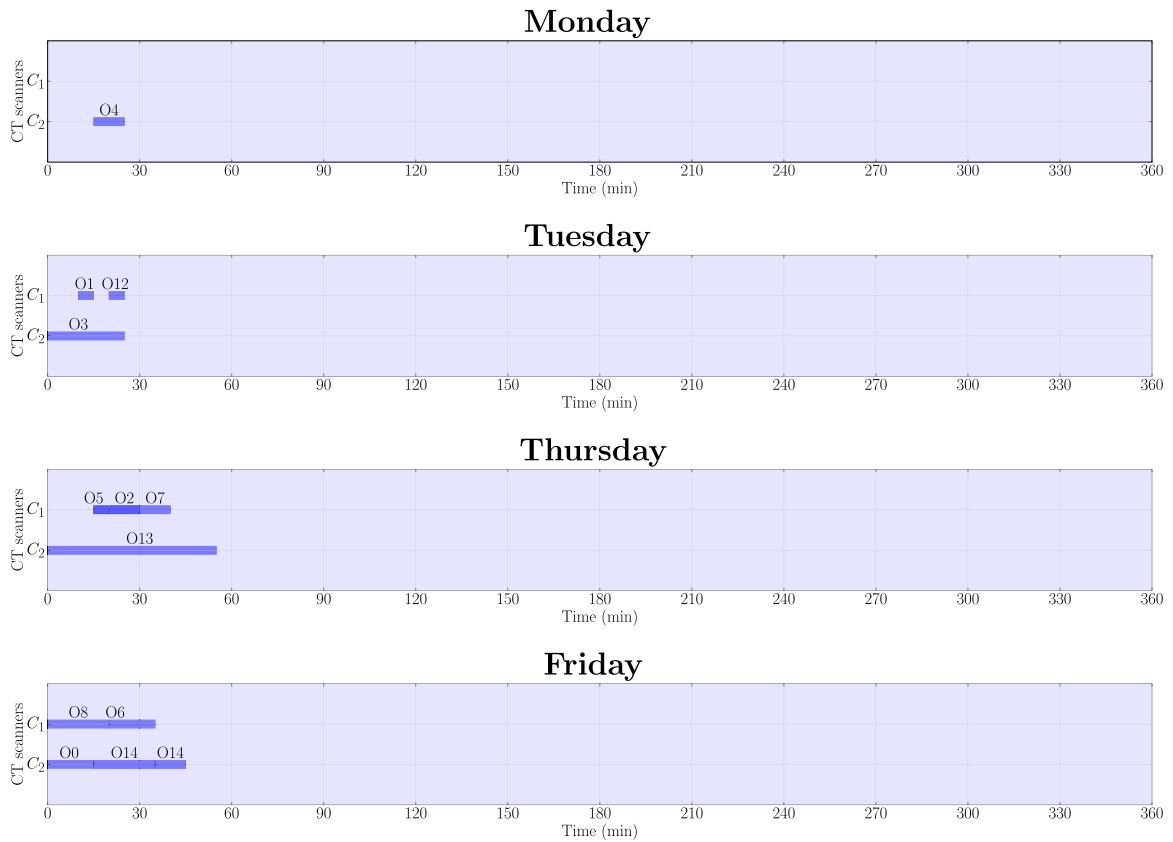
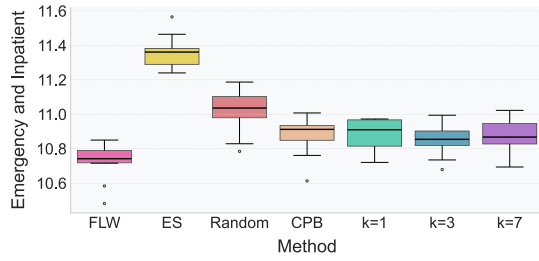


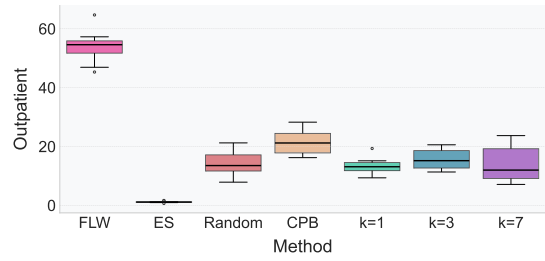
Fig. 10. Context C_1 – Example of schedule generated by NSGA-II with LBA warm start $K = 5$, when the final solution is selected from the Pareto front by considering only the minimization of waiting times of outpatients. We present only the days with scheduled outpatients.

appear even more extreme than those identified through the Pareto-front approximation produced by the genetic algorithm. Although the contribution of the warm-start strategies over random initialization remains appreciable, the behavior of the ES policy suggests that further improvements are still possible for this class of approaches.

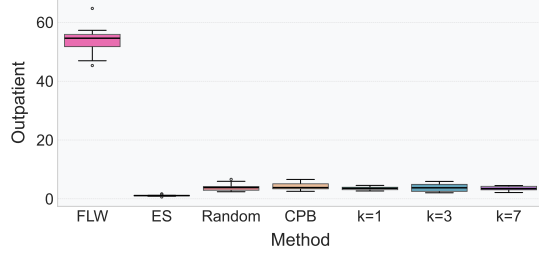
We perform the Wilcoxon signed-rank test to assess the statistical differences between the LBA warm start with $K = 1$ and the two baseline policies, using the fitness values obtained for non-elective patients (emergency patients and inpatients) and outpatients to perform the test. When compared with the FLW policy, the LBA warm start is



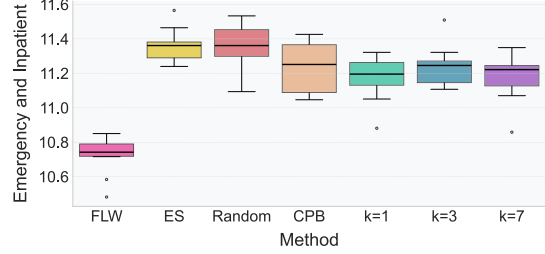
(a) Fitness function values for emergency patients and inpatients when the final solution is selected from the Pareto front by considering only the minimization of completion times for both emergency patients and inpatients.



(b) Fitness function values for outpatients when the final solution is selected from the Pareto front by considering only the minimization of completion times for both emergency patients and inpatients.



(c) Fitness function values for outpatients when the final solution is selected from the Pareto front by considering only the minimization of waiting times of outpatients.



(d) Fitness function values for emergency patients and inpatients when the final solution is selected from the Pareto front by considering only the minimization of waiting times of outpatients.

Fig. 11. Comparison of fitness function values for non-elective patients and outpatients under different prioritization strategies.

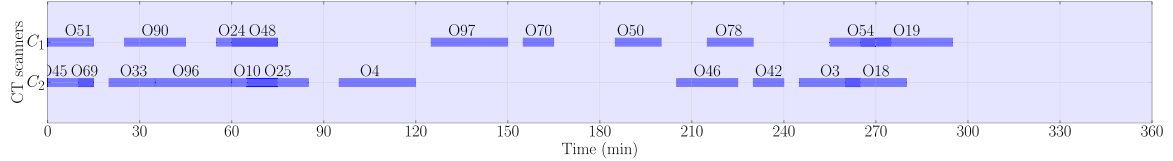


Fig. 12. Context C_2 – Example of schedule (one day, Thursday) generated by NSGA-II with LBA warm start with $K = 1$ for context C_2 , when the final solution is selected from the Pareto front by considering only the minimization of completion times of emergency patients and inpatients.

statistically significantly better when considering outpatient results (p -value of 0.002) and statistically significantly worse when considering non-elective patients (p -value of 0.002). In contrast, compared with the ES policy, the LBA warm start is statistically significantly better with respect to the non-elective patient fitness (p -value of 0.002) and statistically significantly worse with respect to the outpatient fitness (p -value of 0.002).

These results further support the main idea behind the proposed approach. The two baselines are inherently biased toward one of the two objectives: FLW prioritizes non-elective patients, whereas ES prioritizes outpatients. In contrast, our approach generates a diverse set of solutions capable of effectively balancing both objectives simultaneously. Moreover, the extreme solutions identified by our method are statistically superior to the worst-performing baseline in both cases, demonstrating that this improved balance is achieved without sacrificing performance at the extremes.

Fig. 12 shows an example schedule that prioritizes minimizing the completion times of non-elective patients. In this case, outpatients are scheduled mostly at the beginning and end of the shift, thereby freeing peak hours to accommodate non-elective patients. As observed in the real-case context, an overlapping strategy is adopted to reduce the

likelihood that, upon arrival, an emergency patient or an inpatient finds the CT scanner occupied by an outpatient.

Fig. 13 illustrates an example schedule focused on minimizing waiting times for outpatients. Outpatients are distributed throughout the shift, with greater spacing at the beginning and at the end of the shift, while exhibiting a higher degree of overlap during the central time interval.

6.5. Robustness analysis under different prediction models

In this section, we present a further analysis with two main objectives. First, we evaluate and compare the effectiveness of various ML methods employed as regressors to predict outpatient exam duration within our learning-based adaptive approach. Second, considering that the selected ML models demonstrate differing levels of performance according to standard accuracy metrics, we examine the relationship between prediction error and the effectiveness of the prediction-based optimization approach, which provides insight about the robustness of the LBA warm start.

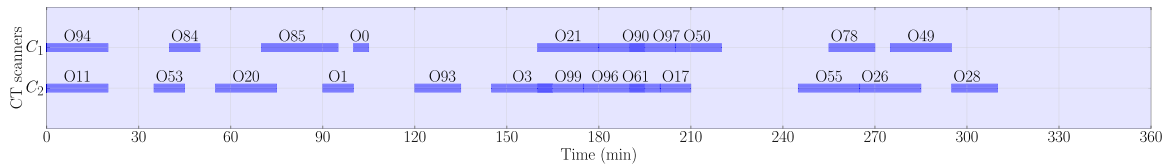


Fig. 13. Context C_2 – Example of schedule (one day, Thursday) generated by NSGA-II with LBA warm start with $K = 1$, when the final solution is selected from the Pareto front by considering only the minimization of waiting times of outpatients.

Table 2

Comparison of performance indicators for predictions (MAE and RMSE expressed in minutes) and for Pareto non-dominated solutions generated by NSGA-II using the three different ML models within the LBA warm start with $K = 1$ for the outpatient-mix context (average values over 10 instances). In the last column (W), we indicate with $\uparrow$, $\downarrow$, or $\times$ the presence of a statistical significance in the comparison with the random initialization with the same ML model. The p -value is reported in parentheses.

ML model	Accuracy			Initialization	Solution quality						
	MAE	RMSE	R^2		#	C	H	Δ	Sp	MID	W (p -value)
MLP	4.07	6.36	0.302	Random	45.0	0.70	707.92	1.12	0.34	1.98	–
				CPB	29.5	0.77	707.79	1.11	0.95	3.51	$\times$ (0.769)
				LBA	63.6	0.94	716.41	1.10	0.22	1.59	$\uparrow$ (0.019)
DT	4.13	6.43	0.238	Random	26.7	0.00	52.20	1.01	0.81	26.95	–
				CPB	41.8	0.40	673.92	1.11	0.49	3.71	$\uparrow$ (0.002)
				LBA	41.8	0.47	686.53	1.05	0.42	3.28	$\uparrow$ (0.002)
CatBoost	4.01	6.35	0.303	Random	29.2	0.16	650.73	1.01	0.32	4.57	–
				CPB	26.2	0.17	648.26	1.09	0.74	5.75	$\times$ (0.625)
				LBA	19.8	0.35	670.96	0.99	0.29	4.15	$\uparrow$ (0.002)

To this aim, we report results for the outpatient-mix context with the proposed LBA warm start, with $K = 1$, which was the best-performing configuration in Section 6.4. This setting is chosen for simplicity and to provide a representative example. The optimization is executed using predictions generated by MLP (i.e., the model used for the computational analysis in Section 6.4), Decision Tree (DT), and Categorical Boosting (CatBoost), both for the fitness function evaluation within evolutionary search and for generating the LBA warm-start solutions. Further details on these ML methods used for regression are provided in Appendix B.

For the comparison of the accuracy, we report a summary of the results of an analysis in which we trained and tested the prediction models. We used BayesSearchCV from the `skopt` library, which optimizes hyperparameters through a cross-validated search over different parameter configurations. We also employed Recursive Feature Elimination using the `RFECV` function from the `sklearn` library to identify the most relevant features influencing model performance. The feature elimination retained all variables listed in Section 6.1, indicating their relevance to the models. For all methods, the predominant factor influencing task duration is the nature of the medical procedure. Additionally, depending on the model, other noteworthy factors include the number of required examinations, department of origin, referring physician, patient age, and patient type.

Notably, the time required to compute predictions after the hyperparameter tuning phase (which only needs to be performed once) is negligible; for example, training the MLP model on our dataset requires approximately 48.3 s, while less than 0.01 s are needed to compute predictions for every new instance.

Results are reported in the first part of Table 2 considering the Mean Absolute Error (MAE), the Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) as accuracy metrics. CatBoost is the most accurate technique, with slightly better results compared to the MLP across all three metrics, and a more significant difference with respect to the DT.

Regarding the quality of the Pareto front, we compute the indicators described in Section 6.2. The Wilcoxon signed-rank test is used to compare the CPB and LBA warm starts with the corresponding random warm starts generated by their respective ML models, in order to assess statistical significance relative to the from-scratch genetic algorithm.

Since the ideal and reference points are recomputed based on the fronts under comparison, the indicator values vary accordingly. In particular, the hypervolume is sensitive to the choice of the reference point. In this case, the poor performance of DT with random initialization results in a distant reference point, which tends to favor more widely spread fronts rather than higher-quality solutions when the fronts do not strictly Pareto-dominate each other. We observe that the algorithm employing MLP-based predictions outperforms the others in terms of coverage, hypervolume, and mean ideal distance. Moreover, we perform an additional series of Wilcoxon signed-rank tests on the hypervolume values to compare the random initialization with MLP-based predictions against all initialization strategies using DT and CatBoost predictions, in addition to the tests reported in Table 2. The results indicate that the MLP-based approach is consistently statistically superior to the other combinations, with a p -value of 0.002. When comparing the three initialization strategies within each ML model, we find that the LBA warm start performs better across almost all metrics and is consistently statistically superior to the random initialization approach. Furthermore, the CPB warm start is significantly better than random initialization only in the case of DT.

These counterintuitive results can be explained by the fact that, although CatBoost and MLP achieve similar performance on accuracy metrics, MLP produces more diverse predictions, enabling a better differentiation among similar instances. In the considered settings, CatBoost identifies three distinct solutions for the LBA warm start across the ten validation instances. In other words, if we select the LBA warm-start instances most similar to each validation instance, there are three unique instances overall to warm start the ten validation instances, as the lists of the most similar instances for each validation instance largely overlap. In comparison, the MLP model identifies six distinct historical instances to warm start the validation ones. This indicates that the discriminative power of the predictive model is more important than small differences in accuracy.

From a robustness perspective, the results suggest that a good level of predictive accuracy is important for obtaining high-quality solutions, as it directly affects the evaluation of the fitness function and, consequently, the performance of NSGA-II. Interestingly, however, there does not appear to be a direct relationship between predictive

accuracy and solution quality, as other factors (e.g., the model's discriminative capability) also play a relevant role. Finally, the results clearly indicate that, even for predictive models with limited accuracy and discriminative power, such as DT, the LBA warm start is able to yield a significant improvement in the performance of the evolutionary search.

6.6. Managerial insights

The Pareto front obtained by the proposed approach provides hospital managers with a set of alternative scheduling policies that explicitly capture the trade-off between outpatient waiting times and the completion times of inpatients and emergency patients. Rather than prescribing a single solution, the framework enables decision-makers to select the most appropriate schedule based on current operational priorities.

Solutions along the Pareto front represent different trade-offs between outpatient waiting times and the completion times of inpatients and emergency patients. At one extreme, outpatient-focused schedules reduce waiting times at the expense of longer completion times for other patients, while the opposite holds when prioritizing inpatient and emergency care. Intermediate solutions provide balanced trade-offs, allowing managers to adapt scheduling decisions to changing operational priorities without recomputing the schedule.

Overall, the results highlight that the integration of the LBA warm start not only improves solution quality but also enhances the practical usability of the Pareto front by providing well-distributed and high-quality alternatives. This enables more informed and transparent decision-making in hospital scheduling, supporting the alignment of operational policies with strategic objectives.

From an operational perspective, the analysis of the Pareto-efficient solutions reveals consistent scheduling patterns that can inform managerial decisions in outpatient appointment systems. In particular, in context C_1 , solutions prioritizing outpatient waiting times tend to schedule appointments at the beginning of the shift, before emergency arrivals. Conversely, solutions that prioritize inpatient and emergency performance concentrate outpatient appointments toward the end of the shift, after the arrival of most emergency and inpatient cases. In context C_2 , solutions prioritizing outpatient waiting times distribute appointments more evenly throughout the shift, thereby reducing congestion and smoothing patient flow. Conversely, solutions that prioritize inpatient and emergency performance concentrate outpatient appointments at the beginning and at the end of the shift, avoiding the central period, during which emergency and inpatient arrivals are more frequent.

Despite its good performance, the LBA warm-start approach presents some practical considerations. Its effectiveness depends on the availability and quality of historical data, and may be influenced by changes in the underlying system over time. Moreover, small values of K may lead to an excessive focus on specific past scenarios, suggesting the need for a careful calibration of this parameter. Finally, performance also depends on the characteristics of the predictive model, not only in terms of accuracy but also of discriminative capability, highlighting opportunities for further improvements through model selection and integration.

7. Conclusions

This study presented a novel stochastic programming approach for outpatient appointment scheduling in the context of emergency radiology, addressing the coexistence of outpatients, inpatients, and emergency patients. By integrating predictive modeling with multi-objective optimization, the proposed methodology balances clinical priorities while coping with operational uncertainty. In general, the introduction of a learning-based adaptive warm-start strategy for NSGA-II enhances improvements compared with a random initialization.

The empirical analysis, conducted on real hospital data and extended to operational contexts with increased outpatient demand, confirmed the robustness and scalability of the approach. Results show that the framework is able to identify schedules that substantially reduce non-elective patients' completion times while providing competitive waiting times for outpatients, thereby supporting a more equitable and efficient use of CT resources.

In the warm-start strategy, the best number K of historical instances to initialize the search varies across the two operational contexts. In both contexts, the best value of K depends on the specific instance and should be selected accordingly to fully exploit the benefits of the proposed approach.

From a managerial perspective, the findings provide actionable insights for hospital administrators. First, the concentration of outpatient appointments in specific time windows reduces the likelihood of interference with non-elective arrivals, thereby ensuring timely treatment for such patients. Furthermore, the overlapping of outpatient appointments emerges as a viable strategy to lower the probability that non-elective patients face delays due to scanner occupation, although it raises concerns about fairness and service quality for outpatients. In addition, the explicit modeling of trade-offs through Pareto fronts supports informed decision-making: managers can select solutions that balance clinical urgency with outpatient experience, depending on institutional priorities and patient mix. Finally, the scalability analysis highlights that the proposed approach remains effective under increased demand, offering a robust tool for planning in contexts of rising imaging needs.

Several directions for future research emerge from this study. First, while the proposed methodology has been applied to the specific case of CT scan scheduling in emergency radiology, the framework is general and can be extended to other outpatient appointment scheduling contexts. Exploring its applicability to different clinical services would broaden its impact. Second, further sources of uncertainty could be incorporated, such as delays in outpatients' arrivals or in the transportation of inpatients from other hospital wards, to better capture real-world variability. Finally, it would be of interest to investigate the interaction between the proposed offline scheduling approach and alternative online allocation strategies, in order to design integrated decision-making systems that combine predictive planning with adaptive real-time adjustments.

CRedit authorship contribution statement

Sara Cambiaghi: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation.
Davide Duma: Writing – original draft, Supervision, Methodology, Conceptualization.

Code availability

The complete Python code of the NSGA-II implementation and the LBA warm-start algorithm is available on GitHub at the following link:
https://github.com/davide-duma/learning_based_adaptive_evolutionary_search

Acknowledgments

The work of the author Sara Cambiaghi is supported by the Ph.D. fellowship funded under the Italian National Recovery and Resilience Plan (PNRR), as part of the program for research doctorates and innovative doctorates for public administration and cultural heritage (D.M. 118, 02-03-2023). The authors wish to thank Ludovico Ambrosi, Chandra Bortolotto, Luisa Carone, Alessio Consonni, and Lorenzo Preda from the Emergency Department of the Radiology Institute, Policlinico San Matteo, Pavia, for their indispensable contribution, helping us understand the managerial aspects of the CT scan service and providing the data.

Table A.3
Glossary of indices, sets, parameters, and variables.

Indices and sets	
$c \in C$	CT scanners
$d, d' \in D$	days
$e, e' \in E$	emergency patients
$i, i' \in I$	inpatients
$o, o' \in O$	outpatients
$p, p' \in P$	general patients
$r \in R$	radiologists
$t, t', t'' \in T, T^-, T^+$	time slots
$\omega \in \Omega$	scenarios
Parameters and random variables	
$a_{o,r}$	1 if radiologist r is skilled for patient o 's exam, 0 otherwise
f	patient's feature vector
$h_{p,\omega}$	transportation time of patient p
M	big-M constant
$\alpha_{p,r,\omega}$	1 if radiologist r is skilled for patient p 's exam, 0 otherwise
$\zeta_{p,\omega}$	exam execution duration of patient p
$\theta_{p,\omega}$	request time of patient p
$\nu_{p,\omega}$	report duration of patient p
ξ	vector of all random variables
$\sigma_{p,p',d,\omega}$	1 if patient p arrives before patient p' on day d , 0 otherwise
$\tau_{p,s,d,\omega}$	1 if request time of patient p is at time s of day d , 0 otherwise
$\gamma_{p,p',\omega}$	1 if patients p, p' belong to the same specialty, 0 otherwise
Decision variables	
$\ell_{p,\omega}$	completion time of patient p
$U_{o,o'}$	1 if outpatients o, o' are scheduled on the same day, 0 otherwise
$U'_{o,o',\omega}$	1 if outpatient o arrives before outpatient o' , 0 otherwise
$W_{o,\omega}$	direct waiting time of outpatient o
$x_{o,t,d,r}$	1 if execution of o scheduled at time t of day d with radiologist r , 0 otherwise
$x'_{p,t,d,r,\omega}$	1 if execution of patient p starts at time t of day d with radiologist r , 0 otherwise
$y_{p,t,d,c,\omega}$	1 if exam of patient p is executed at time t of day d on CT scanner c , 0 otherwise
$z_{p,t,d,r,\omega}$	1 if reporting of patient p is executed at time t of day d by radiologist r , 0 otherwise

Appendix A. Constraints of the second-stage model

We report all parameters, variables, and constraints used to define the search space of the second-stage model with objective function defined in Eq. (14). The model determines the value of the recourse function given the value of first-stage decision variables $\mathbf{x} = (x_{o,t,d,r})$ defined in Eq. (10) and the scenario $\omega \in \Omega$, for which the values of all random variables are realized according to the probability distributions set in Section 3.1, then they are here treated as parameters. All notation is summarized in Table A.3.

Firstly, we define parameters $\tau_{p,\omega}$ representing the time needed to transport the inpatient $p \in I \cup E$ from the hospital ward to the CT-scan waiting room (including any transport time on stretchers or wheelchairs, and by vehicle)

$$\tau_{p,t,d,\omega} = \begin{cases} 1, & \text{if request for exam } p \text{ arrives at the time slot } t \text{ of day } d, \\ 0, & \text{otherwise.} \end{cases}$$

defined for $p \in I \cup E$. Then the arrival time of a non-elective patient p corresponds to the only time slot $t \in T$ such that $\tau_{p,t,d,\omega} = 1$.

Two additional parameters are defined to formulate clearer constraints, which are

$$\sigma_{p,p',d,\omega} = \begin{cases} 1, & \text{if patient } p \text{ arrives before patient } p' \text{ on same day } d, \\ 0, & \text{otherwise,} \end{cases}$$

with $p, p' \in I \cup E$, and

$$\gamma_{p,p',\omega} = \begin{cases} 1, & \text{if exams of patients } p \text{ and } p' \text{ belong to the same specialty,} \\ 0, & \text{otherwise.} \end{cases}$$

with $p, p' \in P$. Furthermore, we define the following auxiliary variables

$$\begin{aligned} x'_{p,t,d,r,\omega} &= \begin{cases} 1, & \text{if the exam of patient } p \text{ starts in slot } t \text{ of day } d \\ & \text{with radiologist } r, \\ 0, & \text{otherwise;} \end{cases} \\ U_{o,o'} &= \begin{cases} 1, & \text{if outpatients } o \text{ and } o' \text{ are scheduled} \\ & \text{on the same day,} \\ 0, & \text{otherwise;} \end{cases} \\ U'_{o,o',\omega} &= \begin{cases} 1, & \text{if outpatient } o \text{ arrives earlier than} \\ & \text{the outpatient } o', \\ 0, & \text{otherwise;} \end{cases} \\ y_{p,t,d,c,\omega} &= \begin{cases} 1, & \text{if the exam of patient } p \text{ takes place in slot } t \text{ of day } d \\ & \text{on CT scanner } c, \\ 0, & \text{otherwise;} \end{cases} \\ z_{p,t,d,r,\omega} &= \begin{cases} 1, & \text{if the report of patient } p \text{ is executed in slot } t \text{ of day } d \\ & \text{by the radiologist } r, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

The first constraints included in the second-stage model ensure that outpatients are examined on the same day and by the radiologist assigned in the first-stage model

$$\sum_{t \in T^+} x'_{o,t,d,r,\omega} = \sum_{t \in T^-} x_{o,t,d,r}, \quad \forall o \in O, \forall d \in D, \forall r \in R, \forall \omega \in \Omega. \quad (\text{A.1})$$

allowing only the allocation of the radiologist in time slots after the outpatient arrival

$$\sum_{t \in T^+, r \in R} t x'_{o,t,d,r,\omega} \geq \sum_{t \in T^-, r \in R} t x_{o,t,d,r}, \quad \forall o \in O, \forall d \in D, \forall \omega \in \Omega. \quad (\text{A.2})$$

The outpatients' direct waiting time is computed accordingly by the constraint

$$w_{o,\omega} \geq \sum_{t \in T^+, d \in D, r \in R} (tx'_{o,t,d,r,\omega}) - \sum_{t \in T^-, d \in D, r \in R} tx_{o,t,d,r,\omega} \quad \forall o \in O, \forall \omega \in \Omega. \quad (A.3)$$

Let us introduce the set C of all CT scanners. Then, the following constraints ensure that each exam is executed properly, which means it is executed on at most one CT scanner at a time

$$\sum_{c \in C} y_{p,t,d,c,\omega} \leq 1, \quad \forall p \in P, \forall t \in T^+, \forall d \in D, \forall \omega \in \Omega, \quad (A.4)$$

and the entire exam execution is performed on the same CT scanner

$$\sum_{t' \in T^+, d' \in D, c' \in C \setminus \{c\}} y_{p,t',d',c',\omega} \leq M(1 - y_{p,t,d,c,\omega}), \quad \forall p \in P, \forall t \in T^+, \forall d \in D, \forall c \in C, \forall \omega \in \Omega, \quad (A.5)$$

where M is a large number to implement the big-M method.

In addition, the CT scanner occupation must be consistent with the exam execution duration

$$\sum_{c \in C, t \in T^+, d \in D} y_{p,t,d,c,\omega} = \zeta_{p,\omega}, \quad \forall p \in P, \forall \omega \in \Omega. \quad (A.6)$$

and the CT scanner and the radiologist must be coordinated at the first slot of this task

$$\sum_{c \in C, t' \in T^+, t \leq t' \leq t + \zeta_{p,\omega} - 1} y_{p,t',d,c,\omega} - \zeta_{p,\omega} \geq M \left(\sum_{r \in R} x'_{p,t,d,r,\omega} - 1 \right), \quad \forall p \in P, \forall t \in T^+, \forall d \in D, \forall \omega \in \Omega. \quad (A.7)$$

Resources must be allocated to at most one task at a time, which means only an exam can be executed on each CT scanner

$$\sum_{p \in P} y_{p,t,d,c,\omega} \leq 1, \quad \forall t \in T^+, \forall d \in D, \forall c \in C, \forall \omega \in \Omega, \quad (A.8)$$

and each radiologist can perform only the execution or the reporting of a single exam

$$\sum_{p \in P} x'_{p,t,d,r,\omega} + \sum_{p \in I \cup E} z_{p,t,d,r,\omega} \leq 1, \quad \forall t \in T^+, \forall d \in D, \forall r \in R, \forall \omega \in \Omega. \quad (A.9)$$

Inpatient and emergency exams must be executed only after the arrival of the patient

$$\sum_{t \in T^+, d \in D, r \in R} tx'_{p,t,d,r,\omega} \geq \theta_{p,\omega} + h_{p,\omega}, \quad \forall p \in I \cup E, \forall \omega \in \Omega, \quad (A.10)$$

and within the current work shift (including tolerated overtime)

$$\sum_{t \in T^+, r \in R} x'_{p,t,d,r,\omega} = \sum_{t \in T} \tau_{p,t,d,\omega}, \quad \forall p \in I \cup E, \forall d \in D, \forall \omega \in \Omega. \quad (A.11)$$

In addition, only radiologists skilled in the exam specialty can be assigned to inpatient and emergency patients: while for outpatients this is ensured by the first-stage model together with constraint (A.2), the other patients are directly managed through the constraints

$$x'_{p,t,d,r,\omega} \leq \alpha_{p,r,\omega}, \quad \forall p \in I \cup E, \forall t \in T^+, \forall r \in R, \forall \omega \in \Omega. \quad (A.12)$$

While constraints (A.1)–(A.12) ensure solutions that reflect the correct execution of CT scans, the next constraints are formulated for the reporting phase, which is modeled only for inpatient and emergency patient because they have an impact on the objective $Q_2(x; \xi(\omega))$. Firstly, the following constraints impose that reports are written only between the end of the exam execution

$$t \sum_{r \in R} z_{p,t,d,r,\omega} \geq t' \sum_{r \in R} x'_{p,t',d,r,\omega} + \zeta_{p,\omega}, \quad \forall p \in I \cup E, t, t' \in T^+, \forall d \in D, \forall \omega \in \Omega, \quad (A.13)$$

and the end of the work shift (including tolerated overtime) for a total number of slots consistent with the report duration

$$\sum_{t \in T^+, r \in R} z_{p,t,d,r,\omega} = v_{p,\omega} \sum_{t \in T} \tau_{p,t,d,\omega}, \quad \forall p \in I \cup E, \forall d \in D, \forall \omega \in \Omega. \quad (A.14)$$

Secondly, the following constraints ensure that reports are written by the same radiologist who is allocated for the CT-scan execution

$$z_{p,t,d,r,\omega} \leq \sum_{t' \in T^+, d' \in D} x'_{p,t',d',r,\omega}, \quad \forall p \in I \cup E, \forall r \in R, \forall \omega \in \Omega. \quad (A.15)$$

ensuring continuity of care. Lastly, the completion time of the report is computed as follows

$$\ell_{p,\omega} \geq t \sum_{r \in R} z_{p,t,d,r,\omega}, \quad \forall p \in I \cup E, \forall t \in T^+, \forall d \in D, \forall \omega \in \Omega. \quad (A.16)$$

The last set of constraints aims at implementing the scheduling and sequencing rules (R1)–(R5). They reflect a reasonable approximation of the current guidelines and policies adopted by the CT-scan service of the case study presented in Section 6.1. The sequencing rule (R4) that defines the dynamic allocation prioritization is implemented by the constraints

$$\sum_{r \in R} x'_{e,t,d,r,\omega} + \left(1 - \sum_{t' \in T} \tau_{e,t',d,\omega} \right) \geq \gamma_{e,p,\omega} \sum_{r \in R, t' \in T^+, \theta_{e,\omega} \leq t' \leq t} x'_{p,t',d,r,\omega}, \quad \forall e \in E, \forall p \in O \cup I, \forall t \in T^+, t \geq \theta_{e,\omega}, \forall d \in D, \forall \omega \in \Omega, \quad (A.17)$$

$$\sum_{r \in R} x'_{i,t,d,r,\omega} + \left(1 - \sum_{t' \in T} \tau_{i,t',d,\omega} \right) \geq \gamma_{i,p,\omega} \sum_{r \in R, t' \in T^+, \theta_{i,\omega} \leq t' \leq t} x'_{o,t',d,r,\omega}, \quad \forall i \in I, \forall o \in O, \forall t \in T^+, t \geq \theta_{i,\omega} + h_{i,\omega}, \forall d \in D, \forall \omega \in \Omega, \quad (A.18)$$

imposing that a patient with a higher priority cannot wait while another exam is starting, unless they belong to different specialties. The former prioritizes emergency exams over inpatient and outpatient exams, while the latter prioritizes inpatients over outpatients.

We implement the sequencing rule (R5), imposing the first-come-first-served rule. It is implemented among emergency patients, inpatients, and outpatients by the constraints

$$\gamma_{e,e',\omega} \sum_{r \in R, t \in T^+} tx'_{e,t,d,r,\omega} \leq \sum_{r \in R, t \in T^+} tx'_{e',t,d,r,\omega} + M(1 - \sigma_{e,e',d,\omega}), \quad \forall e, e' \in E, \forall d \in D, \forall \omega \in \Omega, \quad (A.19)$$

$$\gamma_{i,i',\omega} \sum_{r \in R, t \in T^+} tx'_{i,t,d,r,\omega} \leq \sum_{r \in R, t \in T^+} tx'_{i',t,d,r,\omega} + M(1 - \sigma_{i,i',d,\omega}) \quad \forall i, i' \in I, \forall d \in D, \forall \omega \in \Omega, \quad (A.20)$$

$$\gamma_{o,o',\omega} \sum_{r \in R, t \in T^+} tx'_{o,t,d,r,\omega} \leq \sum_{r \in R, t \in T^+} tx'_{o',t,d,r,\omega} + M(2 - v_{o,o'} - v'_{o,o',\omega}) \quad \forall o, o' \in O, \forall d \in D, \forall \omega \in \Omega, \quad (A.21)$$

where the auxiliary variables $v_{o,o'}$ and $v'_{o,o',\omega}$ are defined by the following constraints

$$v_{o,o'} \geq \sum_{r \in R, t \in T^-} x'_{o,t,d,r,\omega} + \sum_{r \in R, t \in T^-} x'_{o',t,d,r,\omega} - 1, \quad \forall o, o' \in O, d \in D, \forall \omega \in \Omega. \quad (A.22)$$

$$v_{o,o'} \leq 1 - \sum_{r \in R, t \in T^-} x'_{o,t,d,r,\omega} + \sum_{r \in R, t \in T^-, d' \in D, d' \neq d} x'_{o',t,d',r,\omega}, \quad \forall o, o' \in O, d \in D, \forall \omega \in \Omega. \quad (A.23)$$

$$\sum_{r \in R, t \in T^-} tx'_{o,t,d,r,\omega} \geq \sum_{r \in R, t \in T^-} tx'_{o',t,d,r,\omega} - M v'_{o,o',\omega}, \quad \forall o, o' \in O, d \in D, \forall \omega \in \Omega. \quad (A.24)$$

For simplicity, we assume that exactly one radiologist is skilled for each specialty. Under this assumption, rules (R1)–(R3) imply that a patient may wait to start the exam only if at least one required resource (a CT scanner or a skilled radiologist) is unavailable due to

a higher-priority exam. Reporting cannot justify waiting. Accordingly, we introduce the following constraints for outpatients

$$\begin{aligned} \sum_{t'' \in T^-, t' \leq t - \alpha_{p,\omega}} x_{o,t,d,r} &= 1 \wedge \sum_{r \in R} x'_{o,t,d,r,\omega} = 0 \wedge \sum_{r \in R} x'_{o,t',d,r,\omega} = 1 \\ \Rightarrow \sum_{c \in C, p \in P, p \neq o} y_{p,t,d,c,\omega} &= |C| \vee \sum_{p \in P, p \neq o} x'_{p,t,d,r,\omega} = 1, \end{aligned} \quad (A.25)$$

$\forall o \in O, \forall t, t' \in T^+, t' > t, \forall d \in D, \forall r \in R : a_{p,r} = 1, \forall \omega \in \Omega$

inpatients,

$$\begin{aligned} \sum_{r \in R} x'_{i,t,d,r,\omega} &= 0 \wedge \sum_{r \in R} x'_{i,t',d,r,\omega} = 1 \Rightarrow \sum_{c \in C, p \in P, p \neq i} y_{p,t,d,c,\omega} = |C| \vee \\ \sum_{p \in P, p \neq i} x'_{p,t,d,r,\omega} &= 1, \quad \forall i \in I, \forall t, t' \in T^+, \theta_{p,\omega} + h_{p,\omega} \leq t < t', \end{aligned} \quad (A.26)$$

$\forall d \in D, \forall r \in R : \alpha_{p,r,\omega} = 1, \forall \omega \in \Omega,$

and emergency patients

$$\begin{aligned} \sum_{r \in R} x'_{e,t,d,r,\omega} &= 0 \wedge \sum_{r \in R} x'_{e,t',d,r,\omega} = 1 \Rightarrow \sum_{c \in C, p \in P, p \neq e} y_{p,t,d,c,\omega} = |C| \\ \vee \sum_{p \in P, p \neq e} x'_{p,t,d,r,\omega} &= 1, \quad \forall e \in E, \forall t, t' \in T^+, \theta_{p,\omega} \leq t < t', \end{aligned} \quad (A.27)$$

$\forall d \in D, \forall r \in R : \alpha_{p,r,\omega} = 1, \forall \omega \in \Omega.$

These constraints are presented in a nonlinear form for clarity and conciseness. However, they can be reformulated into an equivalent linear form using standard linearization techniques.

Appendix B. Additional information on prediction techniques

The ML techniques used to predict exam durations are briefly described below.

Decision tree. The DT algorithm for regression builds hierarchical decision rules that partition the training data into subsets to minimize a loss function. Starting from a root node, internal nodes evaluate conditions on attributes, and branches represent possible outcomes until leaf nodes provide the final predictions. DTs are widely used in medical applications due to their interpretability (Quinlan, 1986).

Categorical boosting. CatBoost is a boosting method, which combines the predictions of multiple weak learners (typically decision trees) to create a more accurate model. Each weak learner is trained to capture patterns that previous learners missed, using the gradient of the loss function to guide learning. Final predictions are obtained through a weighted sum of the individual learners, aimed at minimizing the overall error (Prokhorenkova et al., 2018).

Multilayer perceptron. An MLP is a neural network composed of interconnected neurons organized in layers. The input receives data, the hidden transform it, and the output produces predictions. During training, connection weights are updated, and activation functions introduce nonlinearity to enable the representation of complex patterns (Rumelhart et al., 1986).

Appendix C. Algorithms' pseudocode

In this section, we present a comprehensive pseudocode of the proposed evolutionary search approach.

Algorithm 1 governs the execution of NSGA-II and requires: the set Ω of scenarios, population size N , number of offspring n_{off} , crossover probability p_c , mutation probability p_m , single-gene crossover probability p_{cg} , single-gene mutation probability p_{mg} , available slots T^+ , working days $|D|$, number k of solutions for the learning-based adaptive warm start, the outpatient waiting list WL , a trained ML regressor $\mathcal{M}$, previously solved instances for warm start, and CPB warm start instances. First, in the prediction phase, $\mathcal{M}$ estimates the duration of each outpatient exam in WL , rounded up to the nearest multiple

of 5 min. Population initialization has three options: random, CPB (loading $CPB_{i=1}^N$), or LBA via `WARMSTARTINIT` (Algorithm 2).

The main loop iterates until n_{off} offspring are generated. Each individual is evaluated using `EVALUATE` (Algorithm 3). Offspring are created via `CROSSOVER` and `MUTATE`, and selection is performed with `SELECTNSGA2` (Section 4.4). The Pareto front of the last offspring constitutes the solution.

Algorithm 2 presents the procedure `WARMSTARTINIT` to constructs the initial population using the learning-based adaptive warm start. Given the current instance, a set of previously solved instances with their Pareto solutions, and the KNN parameter K , we compute the similarity matrix (Section 5.2.1) and select the K nearest instances. For each, we sample N_k solutions (Section 5.2.2) and add them to the warm-start set.

Algorithm 3 defines the evaluation of NSGA-II individuals across all scenarios. For each scenario, the schedule is created starting from the encoding, and the system is simulated day by day and slot by slot. Patients are assigned to available resources according to priority: emergency, inpatient, then outpatient. Patients without available resources are placed in a virtual waiting list and rescheduled in subsequent slots, respecting the priority order.

Waiting times are updated for assigned outpatients, and reporting and completion times are computed for inpatients and emergency patients. The weighted sum of these metrics is accumulated over the week, then the procedure moves to the next scenario. The fitness function is the vector of mean waiting and completion times across all scenarios.

Algorithm 1 NSGA-II for outpatient CT Scheduling

```

1: Input:  $|\Omega|$ ,  $N$ ,  $n_{off}$ ,  $p_c$ ,  $p_m$ ,  $p_{cg}$ ,  $p_{mg}$ ,  $|T^+|$ ,  $|D|$ ,  $k$ ,  $WL = \{x_i\}_{i=1}^{|O|}$ ,  $\mathcal{M}$ 
2:   Previously solved instances  $\{\delta^{(h)}\}_{h \in H}$  and corresponding Pareto
   fronts  $\{Pareto^{(h)}\}_{h \in H}$ 
3:   CPB initialization solutions  $\{CPB\}_{i=1}^N$ 
4: Output: Pareto front of solutions for  $WL$ 
5: Prediction Phase:
6:  $\delta \leftarrow []$ 
7: for  $i = 1$  to  $|O|$  do
8:    $\delta_i \leftarrow \mathcal{M}(x_i)$ , append  $\delta_i$  to  $\delta$  ▷ Predict exam duration
9: end for
10: Initialization of population:
11:  $P \leftarrow \{\}$ 
12: if Random Initialization then
13:   for  $i = 1$  to  $N$  do
14:      $ind \leftarrow [slot_1, \dots, slot_n, day_1, \dots, day_n]$ , add  $ind$  to  $P$  ▷ Random values
15:   end for
16: else if CPB Initialization then
17:    $P \leftarrow \{CPB\}$ 
18: else if Learning Based Warm Start then
19:    $P \leftarrow \text{WARMSTARTINIT}(\delta, \{\delta^{(h)}\}_{h \in H}, \{Pareto^{(h)}\}_{h \in H}, k)$  ▷ In Algorithm 2
20: end if
21: Main Loop:
22: for  $off = 1$  to  $n_{off}$  do
23:   for each individual  $ind$  in  $P$  do
24:      $fitness_{ind} \leftarrow \text{EVALUATE}(ind, S, \delta)$  ▷ In Algorithm 3
25:   end for
26:    $p^{new} \leftarrow \text{CROSSOVER}(parent_1, parent_2, p_{cg}) + \text{MUTATE}(child, p_{mg})$ 
27:    $P \leftarrow \text{SELECTNSGA2}(P \cup p^{new}, N)$ 
28: end for
29:  $ParetoFront \leftarrow \text{EXTRACTPARETOFRONT}(P)$ 
30: return  $ParetoFront$ 

```

Algorithm 2 Learning-Based Adaptive Warm Start

```

1: procedure WARMSTARTINIT( $\delta$ ,  $\{\delta^{(h)}\}_{h \in H}$ ,  $\{Pareto^{(h)}\}_{h \in H}$ ,  $k$ )
2:    $similarities \leftarrow []$ 
3:   for each historical instance  $\delta^{(h)}$  in  $H$  do
4:      $sim^{(h)} \leftarrow \text{COMPUTESIMILARITY}(\delta, \delta^{(h)})$ 
5:     Append  $(h, sim^{(h)})$  to  $similarities$ 
6:   end for
7:   Sort  $similarities$  by  $sim^{(h)}$  in ascending order
8:    $selected \leftarrow$  first  $k$  instances from sorted  $similarities$ 
9:    $P_{ws} \leftarrow \{\}$ 
10:  for each  $(h, sim^{(h)})$  in  $selected$  do
11:     $N_k \leftarrow \text{COMPUTENUMBERSOLUTIONS}(h, k)$ 
12:     $PF^{(h)} \leftarrow$  Pareto front of instance  $\delta^{(h)}$ 
13:     $solutions \leftarrow$  Sample  $N_k$  random solutions from  $PF^{(h)}$ 
14:     $P_{ws} \leftarrow P_{ws} \cup solutions$ 
15:  end for
16:  return  $P_{ws}$ 
17: end procedure

```

Algorithm 3 Fitness Evaluation Procedure

```

1: procedure EVALUATE( $ind$ ,  $S$ ,  $\delta$ )
2:    $results \leftarrow []$ 
3:   for each scenario  $s$  in  $S$  do
4:      $schedule \leftarrow \text{CREATE SCHEDULE}(ind, \delta)$ 
5:      $ct_{emergency} \leftarrow 0$ ,  $wt_{outpatient} \leftarrow 0$ ,  $ct_{inpatient} \leftarrow 0$ 
6:     for  $day = 1$  to  $|D|$  do
7:        $patients_{day} \leftarrow$  Filter  $schedule$  by  $day$ 
8:       for  $slot = 1$  to  $|T^+|$  do
9:         for patient  $p$  in  $patients_{day}$  prioritized by type do
10:          if resources are available for  $p$  at  $slot$  then
11:            Assign  $p$  to CT scanner and physician
12:            Update waiting time weighted by patient priority
13:          end if
14:        end for
15:      end for
16:      Compute reporting times for inpatients and emergency patients
17:      Update  $ct_{emergency}$ ,  $wt_{outpatient}$ ,  $ct_{inpatient}$ 
18:    end for
19:     $f_1 \leftarrow ct_{emergency} + ct_{inpatient}$ ,  $f_2 \leftarrow wt_{outpatient}$ 
20:    Append  $(f_1, f_2)$  to  $results$ 
21:  end for
22:  return  $(\text{mean}(f_1), \text{mean}(f_2))$  across scenarios
23: end procedure

```

Data availability

The data that has been used is confidential.

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